URBAN INSTITUTE DATA VISUALIZATION STYLE GUIDE

Use this data visualization style guide to create a uniform look and feel to all of Urban's charts and graphs. This site contains guidelines that are in line with data visualization best practices and proven design principles. It also eliminates the burden of design and color decisions when creating charts.

These guidelines are primarily for publishing charts to the website. Minor sizing and typographic adjustments need to be made to insert charts into documents.

CHART TYPOGRAPHY

Urban.org uses the font *Lato*. When possible, this font should also be used to create charts. If Lato is not installed on your computer, contact IT. A suitable replacement is Arial. Good chart typography creates a hierarchy among elements and guides the reader through the visual.

	TYPEFACE (ALTERNATIVE)	WEB SIZE	PRINT SIZE	CASE	COLOR	NOTES
FIGURE NUMBER	Lato Regular	11	8	ALL CAPS	#1696d2 or rgb(22, 150, 210)	
Title	Lato Bold (Arial Bold)	18 (24 for maps–see "Maps" section)	12	Title Case	#000000 or rgb(0,0,0)	The main point of the chart. Try to keep shorter than two lines and avoid qualifiers.
Subtitle	Lato Regular (Arial)	14	9.5	Sentence case	#000000 or rgb(0,0,0)	Use this to add qualifiers or further clarification to the title.
X and Y axis titles	Lato Regular Italic (Arial Italic)	12	8.5	Sentence case	#000000 or rgb(0,0,0)	Always horizontal, above the top axis label Include units or multipliers in parenthesis (millions), ($2014)

X and Y axis labels	Lato Regular (Arial)	12	8.5	Sentence case	#000000 or rgb(0,0,0)	Always horizontal, avoid units or multipliers. Those should be added to the axis title in parenthesis
Key labels	Lato Regular (Arial)	12	9.5	Sentence case	#000000 or rgb(0,0,0)	Always horizontal. Avoid redundant key labels if possible.
Direct labels	Lato Bold (Arial Bold)	12	9.5	Sentence case	#000000 or rgb(0,0,0)	Use for line or column charts with three or fewer series.
Data-point label	Lato Regular (Arial)	12	8.5	Sentence case	#000000 or rgb(0,0,0)	Always horizontal. No units or multipliers. Only directly label columns if there are fewer than 10 total columns in the chart.
Source and Notes	Lato Regular (Arial)	11	8	Sentence case	#000000 or rgb(0,0,0)	Bold "**Source:**" and "**Notes:**" as well as any statistical significance indicators.

CHART PARTS

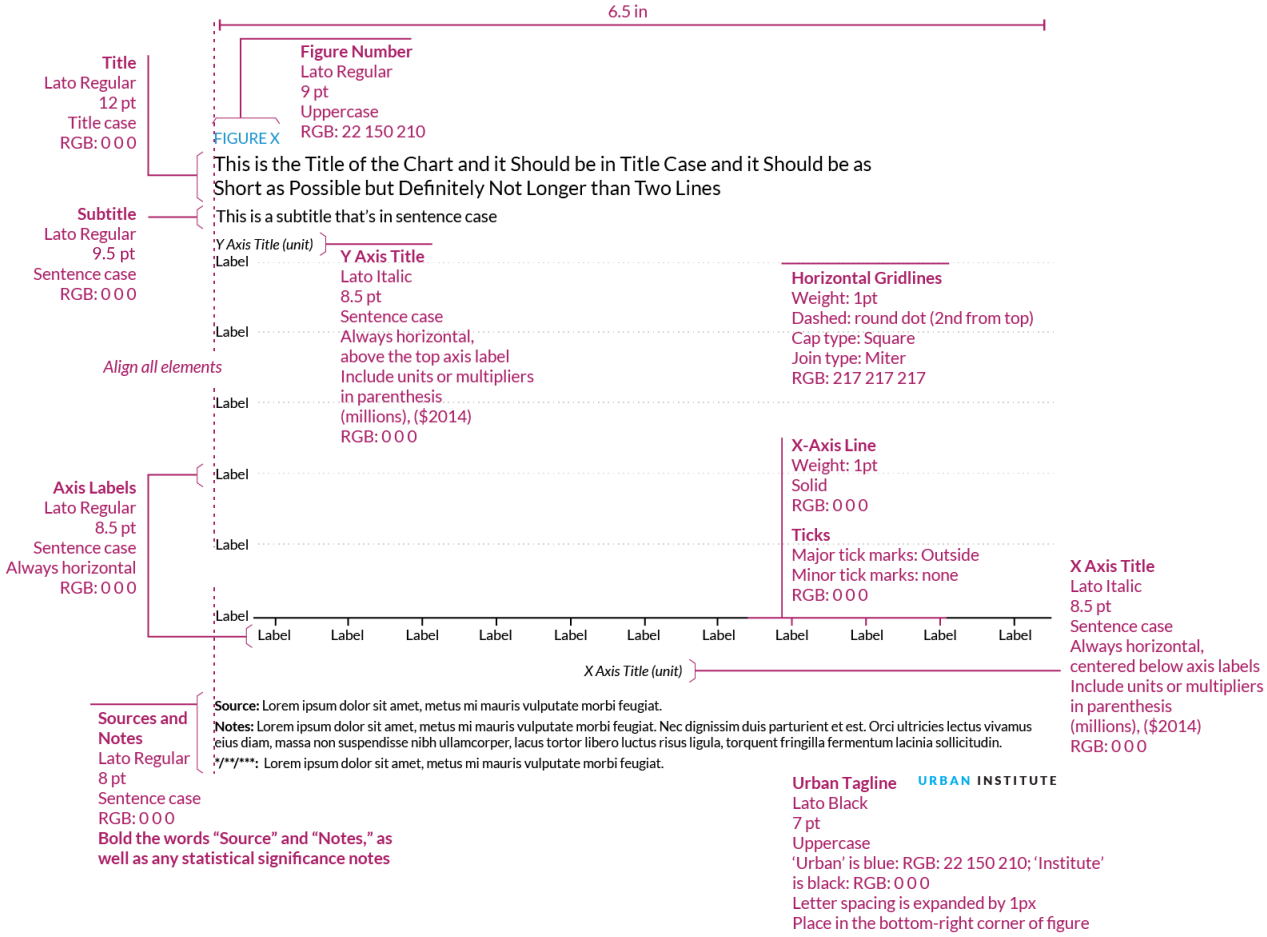
The main content well on the Urban website is 685px wide, which translates to approximately 9.75" wide in Excel. Because Excel's chart title and subtitle fields are limiting in terms of formatting, we use a regular text box for all the text at the top of the graphic, as well as the source and notes text at the bottom. We also include an Urban tag in the lower-right corner of the figure so that when media members copy-paste our charts into their stories, Urban's brand is on them.

This tagline is built from standard text using Lato Regular with 1pt of expanded character spacing. Urban is colored blue—#1696d2 or rgb(22,150,210)—and Institute is black. The text is in uppercase and should be 9pt for web, 7pt for print.

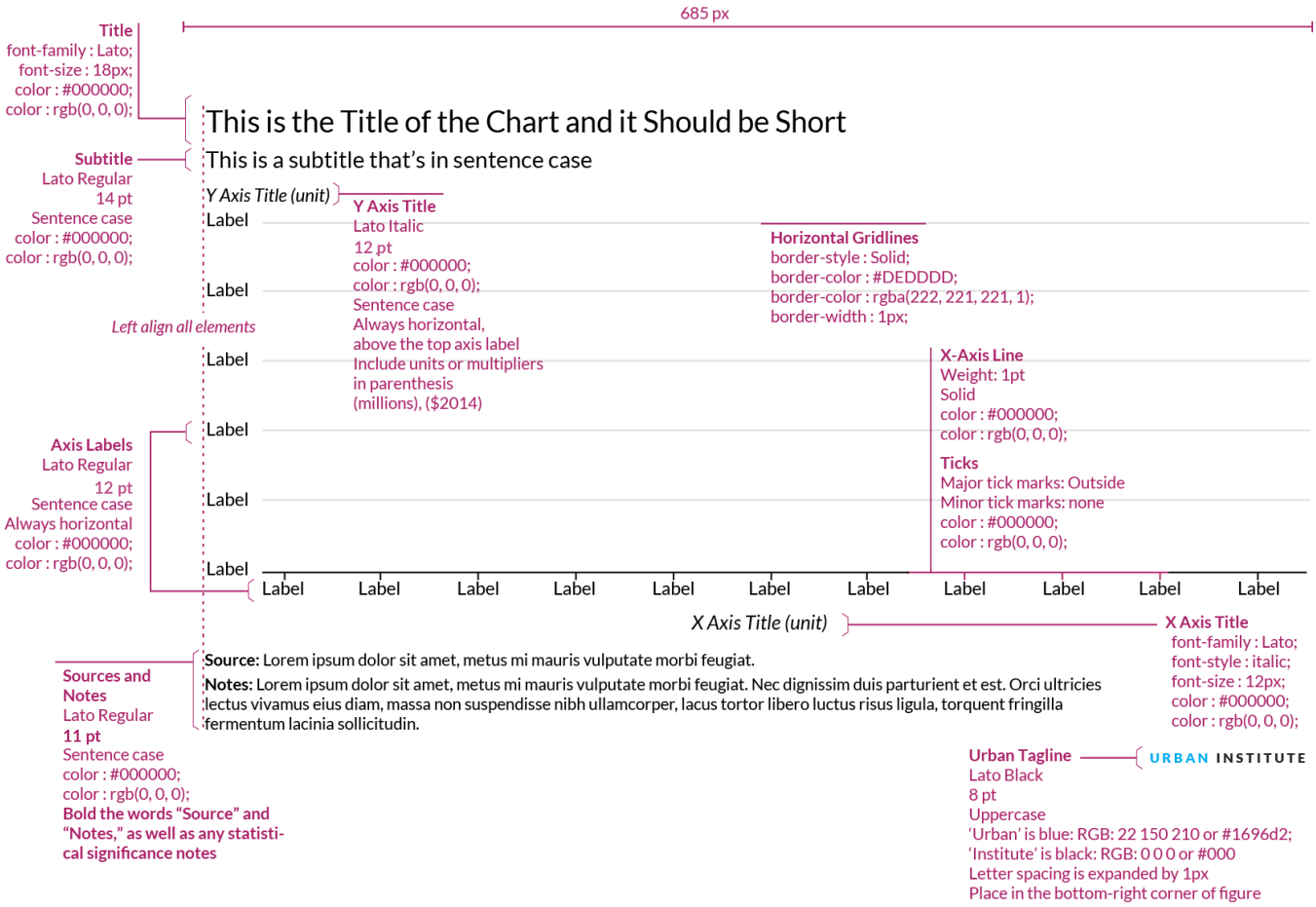
TIPS

- All of Urban's charts will be full-width (685px), so it is important to keep the data density as high as possible. Always include a text reference to your figure to give the data context to the content of the report/brief/blog post. If your chart has only two or three values, consider a couple sentences of text to explain the figure.
- If you find your explanatory sentences do a better job of distilling the information, you might want to consider going without a chart.
- Title: Keep it short and simple. Try to explain the chart in a few words. If you need to add qualifiers (e.g., years, dollars) or further clarification, use a subtitle
- Source and Notes: This is where the technical information about methodology can go. Try to avoid putting this information in the title, labels, or on the chart.
- Legends: Stretch legends across the top of the chart, or to the right. Order them in a logical way, mirroring the order of the data in the charts.

IN PRINT


6.5 in
Title
Lato Regular
12 pt
Title case
RGB: 0 0 0
Figure Number
Lato Regular
9 pt
Uppercase
RGB: 22 150 210
FIGURE X
This is the Title of the Chart and it Should be in Title Case and it Should be as Short as Possible but Definitely Not Longer than Two Lines
Subtitle
Lato Regular
9.5 pt
Sentence case
RGB: 0 0 0
This is a subtitle that's in sentence case
Y Axis Title (unit)
Y Axis Title
Lato Italic
8.5 pt
Sentence case
Always horizontal,
above the top axis label
Include units or multipliers
in parenthesis
(millions), ($2014)
RGB: 0 0 0
Horizontal Gridlines
Weight: 1pt
Dashed: round dot (2nd from top)
Cap type: Square
Join type: Miter
RGB: 217 217 217
Align all elements
Axis Labels
Lato Regular
8.5 pt
Sentence case
Always horizontal
RGB: 0 0 0
X-Axis Line
Weight: 1pt
Solid
RGB: 0 0 0
Ticks
Major tick marks: Outside
Minor tick marks: none
RGB: 0 0 0
X Axis Title
Lato Italic
8.5 pt
Sentence case
Always horizontal,
centered below axis labels
Include units or multipliers
in parenthesis
(millions), ($2014)
RGB: 0 0 0
Label
Label Label Label Label Label Label Label Label Label Label Label
X Axis Title (unit)
Sources and Notes
Lato Regular
8 pt
Sentence case
RGB: 0 0 0
Bold the words "Source" and "Notes," as well as any statistical significance notes
Source: Lorem ipsum dolor sit amet, metus mi mauris vulputate morbi feugiat.
Notes: Lorem ipsum dolor sit amet, metus mi mauris vulputate morbi feugiat. Nec dignissim duis parturient et est. Orci ultricies lectus vivamus eius diam, massa non suspendisse nibh ullamcorper, lacus tortor libero luctus risus ligula, torquent fringilla fermentum lacinia sollicitudin.
*/**/***: Lorem ipsum dolor sit amet, metus mi mauris vulputate morbi feugiat.
Urban Tagline
Lato Black
7 pt
Uppercase
'Urban' is blue: RGB: 22 150 210; 'Institute' is black: RGB: 0 0 0
Letter spacing is expanded by 1px
Place in the bottom-right corner of figure
URBAN INSTITUTE

ON THE WEB


685 px
Title
font-family : Lato;
font-size : 18px;
color : #000000;
color : rgb(0, 0, 0);
This is the Title of the Chart and it Should be Short
Subtitle
Lato Regular
14 pt
Sentence case
color : #000000;
color : rgb(0, 0, 0);
This is a subtitle that's in sentence case
Y Axis Title (unit)
Y Axis Title
Lato Italic
12 pt
color : #000000;
color : rgb(0, 0, 0);
Sentence case
Always horizontal,
above the top axis label
Include units or multipliers
in parenthesis
(millions), ($2014)
Horizontal Gridlines
border-style : Solid;
border-color : #DEDDDD;
border-color : rgba(222, 221, 221, 1);
border-width : 1px;
Left align all elements
Axis Labels
Lato Regular
12 pt
Sentence case
Always horizontal
color : #000000;
color : rgb(0, 0, 0);
X-Axis Line
Weight: 1pt
Solid
color : #000000;
color : rgb(0, 0, 0);
Ticks
Major tick marks: Outside
Minor tick marks: none
color : #000000;
color : rgb(0, 0, 0);
Label
Label Label Label Label Label Label Label Label Label Label Label
X Axis Title (unit)
X Axis Title
font-family : Lato;
font-style : italic;
font-size : 12px;
color : #000000;
color : rgb(0, 0, 0);
Sources and Notes
Lato Regular
11 pt
Sentence case
color : #000000;
color : rgb(0, 0, 0);
Bold the words "Source" and "Notes," as well as any statistical significance notes
Source: Lorem ipsum dolor sit amet, metus mi mauris vulputate morbi feugiat.
Notes: Lorem ipsum dolor sit amet, metus mi mauris vulputate morbi feugiat. Nec dignissim duis parturient et est. Orci ultricies lectus vivamus eius diam, massa non suspendisse nibh ullamcorper, lacus tortor libero luctus risus ligula, torquent fringilla fermentum lacinia sollicitudin.
Urban Tagline
Lato Black
8 pt
Uppercase
'Urban' is blue: RGB: 22 150 210 or #1696d2;
'Institute' is black: RGB: 0 0 0 or #000
Letter spacing is expanded by 1px
Place in the bottom-right corner of figure
URBAN INSTITUTE

COLOR

Urban's main colors are cyan, gray, and black. Yellow and magenta are used as secondary colors throughout the new Urban brand. Teriary colors for graphics include space gray, green, and red, and should be used infrequently.

When selecting colors for charts and graphs, we must first consider the type of data we are presenting. Usually, data can be grouped into one of two main groups: categorical or sequential.

Categorical palettes are best when you want to distinguish discrete chunks of data that do not have an inherent ordering.

Sequential color mapping is appropriate when data range from relatively low or uninteresting values to relatively high or interesting values. For sequential data, it's better to use a palette that has a relatively subtle shift in hue accompanied by a large shift in brightness and saturation. This approach naturally draws the eye to the relatively important parts of the data. Sequential colors are used for sequential groups (not necessarily the way the data trends). So age groups young to old and date ranges might be something that I use sequential colors for.

Note: Yellow and Magenta should be used sparingly, as these colors should be considered for highlighting purposes (such as drawing attention to a certain category or indicating a trend line).

For more information about the subtleties of color, refer to this collection of blog posts (http://blog.visual.ly/?s=%22subtleties+of+color%22) from Visual.ly (http://visual.ly/).

The following color combinations are intended to take some of the guess work out of the process of assigning colors to charts. They're only examples, and can be mixed-and-matched depending on the story you are trying to tell with your data.

When comparing data by gender do not use blue to represent men and pink to represent women. Instead, choose a color combination like yellow and cyan , or another combination of our main graphic colors.

COLOR COMBINATIONS

MAIN GRAPHIC COLORS

Hex: #1696d2 rgb(22,150,210) CMYK: 90,29,0,18	Hex: #d2d2d2 rgb(210,210,210) CMYK: 0,0,0,18	Hex: #000000 rgb(0,0,0) CMYK: 0,0,0,100	Hex: #fdbf11 rgb(253,191,17) CMYK: 0,25,93,1	Hex: #ec008b rgb(236,0,139) CMYK: 0,100,41,7	Hex: #55b748 rgb(85,183,72) CMYK: 54,0,61,28	Hex: #5c5859 rgb(92,88,89) CMYK: 0,4,3,64	Hex: #db2b27 rgb(219,43,39) CMYK: 0,80,82,14

SHADES OF MAIN COLORS

Hex: #cfe8f3	Hex: #a2d4ec	Hex: #73bfe2	Hex: #46abdb	Hex: #1696d2	Hex: #12719e	Hex: #0a4c6a	Hex: #062635
rgb(207,232,243)	rgb(162,212,236)	rgb(115,191,226)	rgb(70,171,219)	rgb(22,150,210)	rgb(18,113,158)	rgb(10,76,106)	rgb(6,38,53)
CMYK: 15,5,0,5	CMYK: 31,10,0,7	CMYK: 49,15,0,11	CMYK: 68,22,0,14	CMYK: 90,29,0,18	CMYK: 89,28,0,38	CMYK: 91,28,0,58	CMYK: 89,28,0,79

var colors=["#CFE8F3","#A2D4EC","#73BFE2","#46ABDB","#1696D2","#12719E","#0A4C6A","#062635"];

Hex: #f5f5f5	Hex: #ececec	Hex: #e3e3e3	Hex: #dcdbdb	Hex: #d2d2d2	Hex: #9d9d9d	Hex: #696969	Hex: #353535
rgb(245,245,245)	rgb(236,236,236)	rgb(227,227,227)	rgb(220,219,219)	rgb(210,210,210)	rgb(157,157,157)	rgb(105,105,105)	rgb(53,53,53)
CMYK: 0,0,0,4	CMYK: 0,0,0,7	CMYK: 0,0,0,11	CMYK: 0,0,0,14	CMYK: 0,0,0,18	CMYK: 0,0,0,38	CMYK: 0,0,0,59	CMYK: 0,0,0,79

var colors=["#F5F5F5","#ECECEC","#E3E3E3","#DCDBDB","#D2D2D2","#9D9D9D","#696969","#353535"];

Hex: #fff2cf	Hex: #fce39e	Hex: #fdd870	Hex: #fccb41	Hex: #fdbf11	Hex: #e88e2d	Hex: #ca5800	Hex: #843215
rgb(255,242,207)	rgb(252,227,158)	rgb(253,216,112)	rgb(252,203,65)	rgb(253,191,17)	rgb(232,142,45)	rgb(202,88,0)	rgb(132,50,21)
CMYK: 0,5,19,0	CMYK: 0,10,37,1	CMYK: 0,15,56,1	CMYK: 0,19,74,1	CMYK: 0,25,93,1	CMYK: 0,39,81,9	CMYK: 0,56,100,21	CMYK: 0,62,84,48

var colors=["#FFF2CF","#FCE39E","#FDD870","#FCCB41","#FDBF11","#E88E2D","#CA5800","#843215"];

Hex: #f5cbdf	Hex: #eb99c2	Hex: #e46aa7	Hex: #e54096	Hex: #ec008b	Hex: #af1f6b	Hex: #761548	Hex: #351123
rgb(245,203,223)	rgb(235,153,194)	rgb(228,106,167)	rgb(229,64,150)	rgb(236,0,139)	rgb(175,31,107)	rgb(118,21,72)	rgb(53,17,35)
CMYK: 0,17,9,4	CMYK: 0,35,17,8	CMYK: 0,54,27,11	CMYK: 0,72,34,10	CMYK: 0,100,41,7	CMYK: 0,82,39,31	CMYK: 0,82,39,54	CMYK: 0,68,34,79

var colors=["#F5CBDF","#EB99C2","#E46AA7","#E54096","#EC008B","#AF1F6B","#761548","#351123"];

Hex: #dcedd9	Hex: #bcdeb4	Hex: #98cf90	Hex: #78c26d	Hex: #55b748	Hex: #408941	Hex: #2c5c2d	Hex: #1a2e19
rgb(220,237,217)	rgb(188,222,180)	rgb(152,207,144)	rgb(120,194,109)	rgb(85,183,72)	rgb(64,137,65)	rgb(44,92,45)	rgb(26,46,25)
CMYK: 7,0,8,7	CMYK: 15,0,19,13	CMYK: 27,0,30,19	CMYK: 38,0,44,24	CMYK: 54,0,61,28	CMYK: 53,0,53,46	CMYK: 52,0,51,64	CMYK: 43,0,46,82

var colors=["#DCEDD9","#BCDEB4","#98CF90","#78C26D","#55B748","#408941","#2C5C2D","#1A2E19"];

Hex: #d5d5d4	Hex: #adabac	Hex: #848081	Hex: #5c5859	Hex: #332d2f	Hex: #262223	Hex: #1a1717	Hex: #0e0c0d
rgb(213,213,212)	rgb(173,171,172)	rgb(132,128,129)	rgb(92,88,89)	rgb(51,45,47)	rgb(38,34,35)	rgb(26,23,23)	rgb(14,12,13)
CMYK: 0,0,0,16	CMYK: 0,1,1,32	CMYK: 0,3,2,48	CMYK: 0,4,3,64	CMYK: 0,12,8,80	CMYK: 0,11,8,85	CMYK: 0,12,12,90	CMYK: 0,14,7,95

var colors=
["#D5D5D4","#ADABAC","#848081","#5C5859","#332D2F","#262223","#1A1717","#0E0C0D"];

Hex: #f8d5d4	Hex: #f1aaa9	Hex: #e9807d	Hex: #e25552	Hex: #db2b27	Hex: #a4201d	Hex: #6e1614	Hex: #370b0a
rgb(248,213,212)	rgb(241,170,169)	rgb(233,128,125)	rgb(226,85,82)	rgb(219,43,39)	rgb(164,32,29)	rgb(110,22,20)	rgb(55,11,10)
CMYK: 0,14,15,3	CMYK: 0,29,30,5	CMYK: 0,45,46,9	CMYK: 0,62,64,11	CMYK: 0,80,82,14	CMYK: 0,80,82,36	CMYK: 0,80,82,57	CMYK: 0,80,82,78

var colors=["#F8D5D4","#F1AAA9","#E9807D","#E25552","#DB2B27","#A4201D","#6E1614","#370B0A"];

ONE GROUP

For one color group, grey should not be used as it loses focus while magenta draws attention

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Hex: #000000
rgb(0,0,0)
CMYK: 0,0,0,100

TWO GROUPS

For two color groups, grey should not used as it loses focus while magenta draws attention

CATEGORICAL

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Hex: #000000
rgb(0,0,0)
CMYK: 0,0,0,100

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Hex: #fdbf11
rgb(253,191,17)
CMYK: 0,25,93,1

Use blue and red sparingly, usually only when comparing political data.

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Hex: #db2b27
rgb(219,43,39)
CMYK: 0,80,82,14

SEQUENTIAL

Hex: #a2d4ec
rgb(162,212,236)
CMYK: 31,10,0,7

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

THREE GROUPS

For three color groups, grey should not used as it loses focus while magenta draws attention

CATEGORICAL

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Hex: #fdbf11
rgb(253,191,17)
CMYK: 0,25,93,1

Hex: #000000
rgb(0,0,0)
CMYK: 0,0,0,100

SEQUENTIAL

Hex: #a2d4ec
rgb(162,212,236)
CMYK: 31,10,0,7

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Hex: #0a4c6a
rgb(10,76,106)
CMYK: 91,28,0,58

FOUR GROUPS

CATEGORICAL

Hex: #000000
rgb(0,0,0)
CMYK: 0,0,0,100

Hex: #d2d2d2
rgb(210,210,210)
CMYK: 0,0,0,18

Hex: #fdbf11
rgb(253,191,17)
CMYK: 0,25,93,1

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Use yellow and magenta sparingly, usually only for highlighting data.

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Hex: #ec008b
rgb(236,0,139)
CMYK: 0,100,41,7

Hex: #fdbf11
rgb(253,191,17)
CMYK: 0,25,93,1

Hex: #5c5859
rgb(92,88,89)
CMYK: 0,4,3,64

SEQUENTIAL

Hex: #cfe8f3
rgb(207,232,243)
CMYK: 15,5,0,5

Hex: #73bfe2
rgb(115,191,226)
CMYK: 49,15,0,11

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Hex: #0a4c6a
rgb(10,76,106)
CMYK: 91,28,0,58

FIVE GROUPS

CATEGORICAL

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Hex: #fdbf11
rgb(253,191,17)
CMYK: 0,25,93,1

Hex: #d2d2d2
rgb(210,210,210)
CMYK: 0,0,0,18

Hex: #ec008b
rgb(236,0,139)
CMYK: 0,100,41,7

Hex: #000000
rgb(0,0,0)
CMYK: 0,0,0,100

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18

Hex: #0a4c6a
rgb(10,76,106)
CMYK: 91,28,0,58

Hex: #d2d2d2
rgb(210,210,210)
CMYK: 0,0,0,18

Hex: #fdbf11
rgb(253,191,17)
CMYK: 0,25,93,1

Hex: #332d2f
rgb(51,45,47)
CMYK: 0,12,8,80

SEQUENTIAL

Hex: #cfe8f3
rgb(207,232,243)
CMYK: 15,5,0,5
	Hex: #73bfe2
	rgb(115,191,226)
	CMYK: 49,15,0,11
		Hex: #1696d2
		rgb(22,150,210)
		CMYK: 90,29,0,18
			Hex: #0a4c6a
			rgb(10,76,106)
			CMYK: 91,28,0,58
				Hex: #000000
				rgb(0,0,0)
				CMYK: 0,0,0,100

SIX GROUPS

CATEGORICAL

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18
	Hex: #ec008b
	rgb(236,0,139)
	CMYK: 0,100,41,7
		Hex: #fdbf11
		rgb(253,191,17)
		CMYK: 0,25,93,1
			Hex: #000000
			rgb(0,0,0)
			CMYK: 0,0,0,100
				Hex: #d2d2d2
				rgb(210,210,210)
				CMYK: 0,0,0,18
					Hex: #55b748
					rgb(85,183,72)
					CMYK: 54,0,61,28

Hex: #1696d2
rgb(22,150,210)
CMYK: 90,29,0,18
	Hex: #d2d2d2
	rgb(210,210,210)
	CMYK: 0,0,0,18
		Hex: #ec008b
		rgb(236,0,139)
		CMYK: 0,100,41,7
			Hex: #fdbf11
			rgb(253,191,17)
			CMYK: 0,25,93,1
				Hex: #332d2f
				rgb(51,45,47)
				CMYK: 0,12,8,80
					Hex: #0a4c6a
					rgb(10,76,106)
					CMYK: 91,28,0,58

SEQUENTIAL

Hex: #cfe8f3
rgb(207,232,243)
CMYK: 15,5,0,5
	Hex: #a2d4ec
	rgb(162,212,236)
	CMYK: 31,10,0,7
		Hex: #73bfe2
		rgb(115,191,226)
		CMYK: 49,15,0,11
			Hex: #46abdb
			rgb(70,171,219)
			CMYK: 68,22,0,14
				Hex: #1696d2
				rgb(22,150,210)
				CMYK: 90,29,0,18
					Hex: #12719e
					rgb(18,113,158)
					CMYK: 89,28,0,38

SEVEN GROUPS

CONSIDER CONSOLIDATING CATEGORIES OR BREAKING UP THE CHART. SEVEN GROUPS IS REALLY TOO MANY.

EXAMPLES

The following examples use the styles and colors to illustrate common chart types. To deconstruct any example, download the the master Excel file.

EXCEL EXAMPLE FILES

WEB VERSION (chart-styles.xlsx) PRINT VERSION (chart-styles-pubs.xlsx)

BAR AND COLUMN CHARTS

WHEN TO USE A BAR OR COLUMN CHART

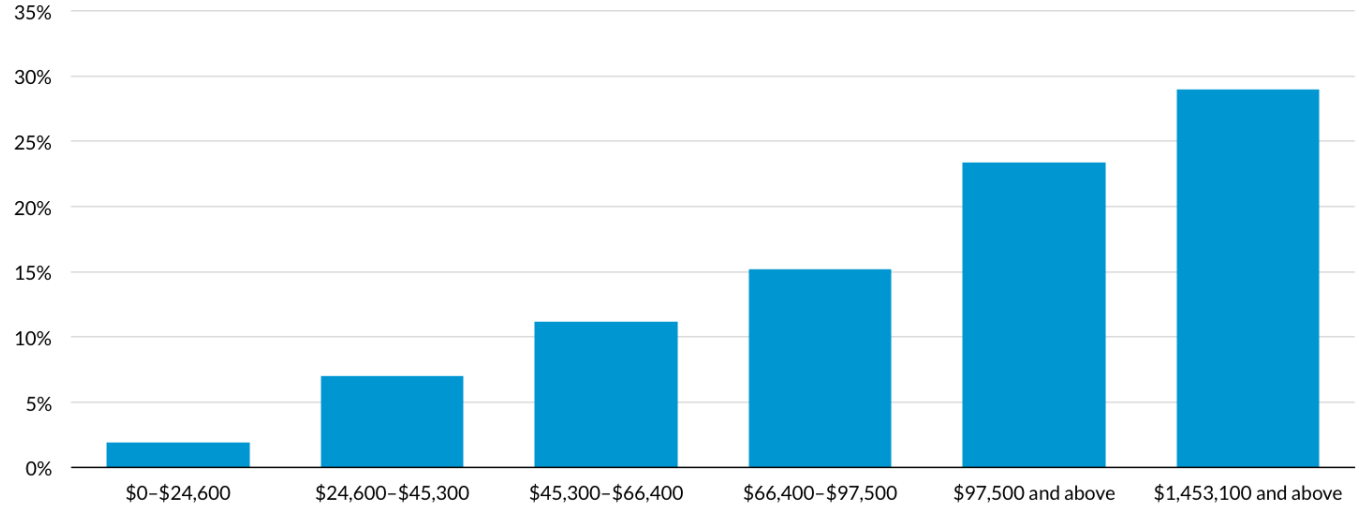
- For mostly for one variable
- To compares numerical values for different observations
- To show relative amounts
- To break one numerical variable out into different subgroups with grouped or stacked bars or columns

STYLE TIPS

- Avoid 3D bars or columns
- The y-axis should start at zero (there are a few instances when it is okay for the y-axis not to start at zero). (http://qz.com/418083/its-ok-not-to-start-your-y-axis-at-zero/)
- The width of the bars should be about twice the width of the space between the bars.
- If all the bars measure the same variable, make them all the same color. Different shades have no relevance to the data.
- If you are showing fewer than 10 bars, consider eliminating the horizontal gridlines and y-axis line and directly labeling the data points.
- To differentiate subsets of data, projections, or averages, consider using a different color shade or gray.
- Legends should be stretched across the top of the chart and the order should match the order in the chart.
- Sequential series should be shaded from lightest to darkest for easy comparison.

Title in 18pt Title Case
Subtitle in 14pt sentence case

Units if needed (thousands of dollars)

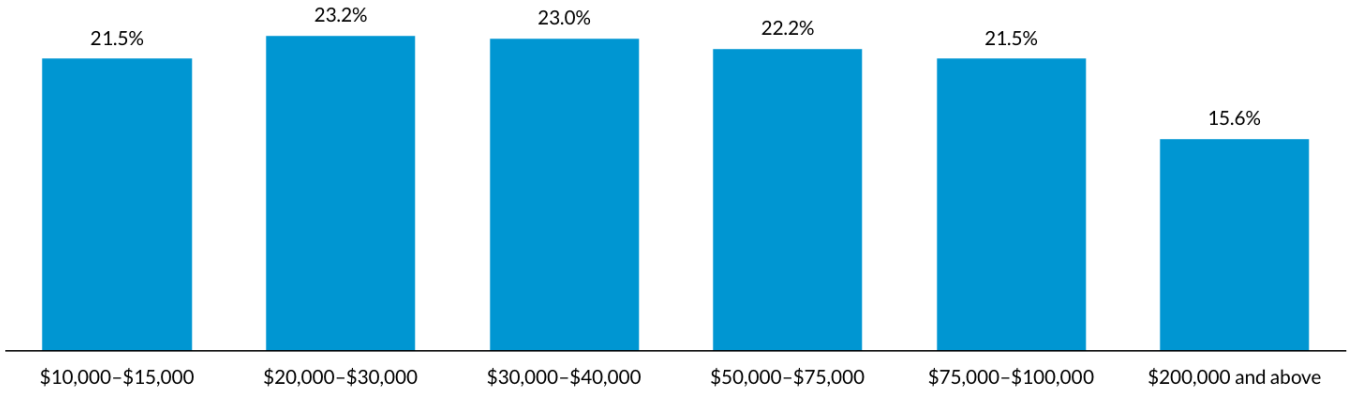


Source: This is the source of the chart.
Notes: This is the notes section of the chart.

URBAN INSTITUTE

Title in 18pt Title Case
Subtitle in 14pt sentence case

Units if needed (thousands of dollars)



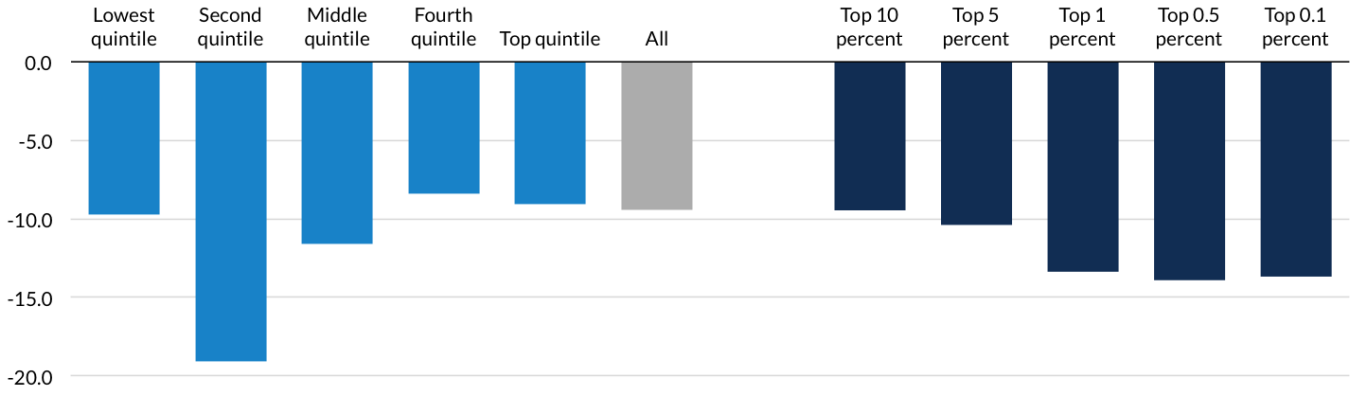
Source: This is the source of the chart.
Notes: This is the notes section of the chart.

URBAN INSTITUTE

Title in 18pt Title Case

Subtitle in 14pt sentence case

Units if needed (thousands of dollars)



Source: This is the source of the chart.
Notes: This is the notes section of the chart.

URBAN INSTITUTE

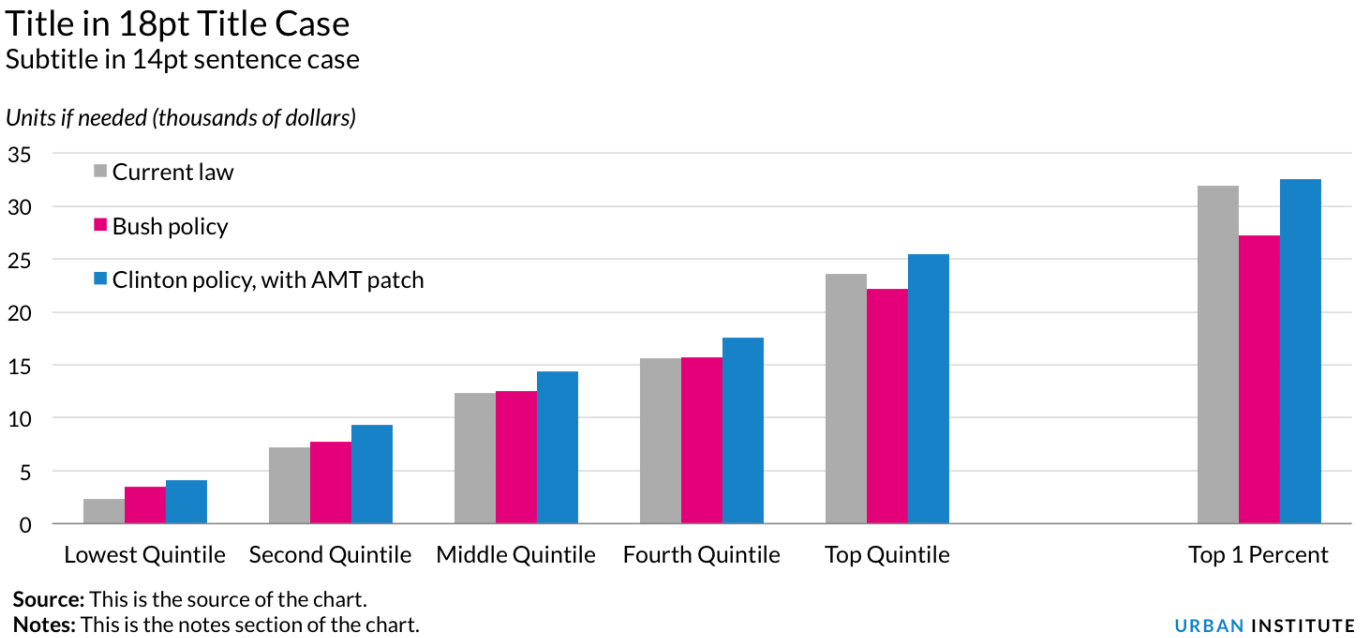
NOTES ABOUT THIS CHART

- Columns on the right in the "Top 10 percent" are a lighter sequential shade to differentiate them from the quintiles.
- The "All" category is shaded gray to create visual contrast from the rest of the columns.
- This chart isn't perfect because the columns on the right side of the chart are disproportionately large—their categories are a fraction of the others, yet visually they are equally weighted. A possible solution is to use conditional formatting (https://support.office.com/en-nz/article/Quick-start-Apply-conditional-formatting-6b6f7c2a-5d62-45a1-8f67-584a76776d67) and build on a traditional distributional table.

Distribution of Federal Tax Change by Cash Income Percentile, 2010

Cash Income Percentile	Percent Change in After-Tax Income	Share of Total Federal Tax Change	Average Federal Tax Change		Share of Federal Taxes	
			Dollars	Percent	Change (% Points)	Under the Proposal
Lowest quintile	0.4	0.4	-37	-9.8	0.0	0.4
Second quintile	2.1	5.2	-437	-19.1	-0.3	2.3
Middle quintile	2.3	9.7	-814	-11.6	-0.2	7.7
Fourth quintile	2.3	15.5	-1,305	-8.4	0.2	17.6
Top quintile	3.6	68.9	-5,809	-9.1	0.3	71.9
All	3.0	100.0	-1,686	-9.4	0.0	100.0
Addendum						
Top 10 percent	4.0	56.1	-9,457	-9.5	0.0	55.9
Top 5 percent	4.7	48.1	-16,223	-10.4	-0.5	43.2
Top 1 percent	6.7	36.8	-62,007	-13.4	-1.1	24.8
Top 0.5 percent	7.3	31.1	-104,678	-13.9	-1.0	20.0
Top 0.1 percent	7.8	18.6	-314,150	-13.7	-0.6	12.3

Source: Urban-Brookings Tax Policy Center Microsimulation Model (version 1006-1).



NOTES ABOUT THIS CHART

- When dealing with political comparisons, lean on the identifiable colors of the parties. Red colors are generally associated with Republicans and blues are generally associated with Democrats.
- When working on a chart that has political categories, be sure to use a nonpolitical color for baseline or unaffiliated series (in this case, “current law”).

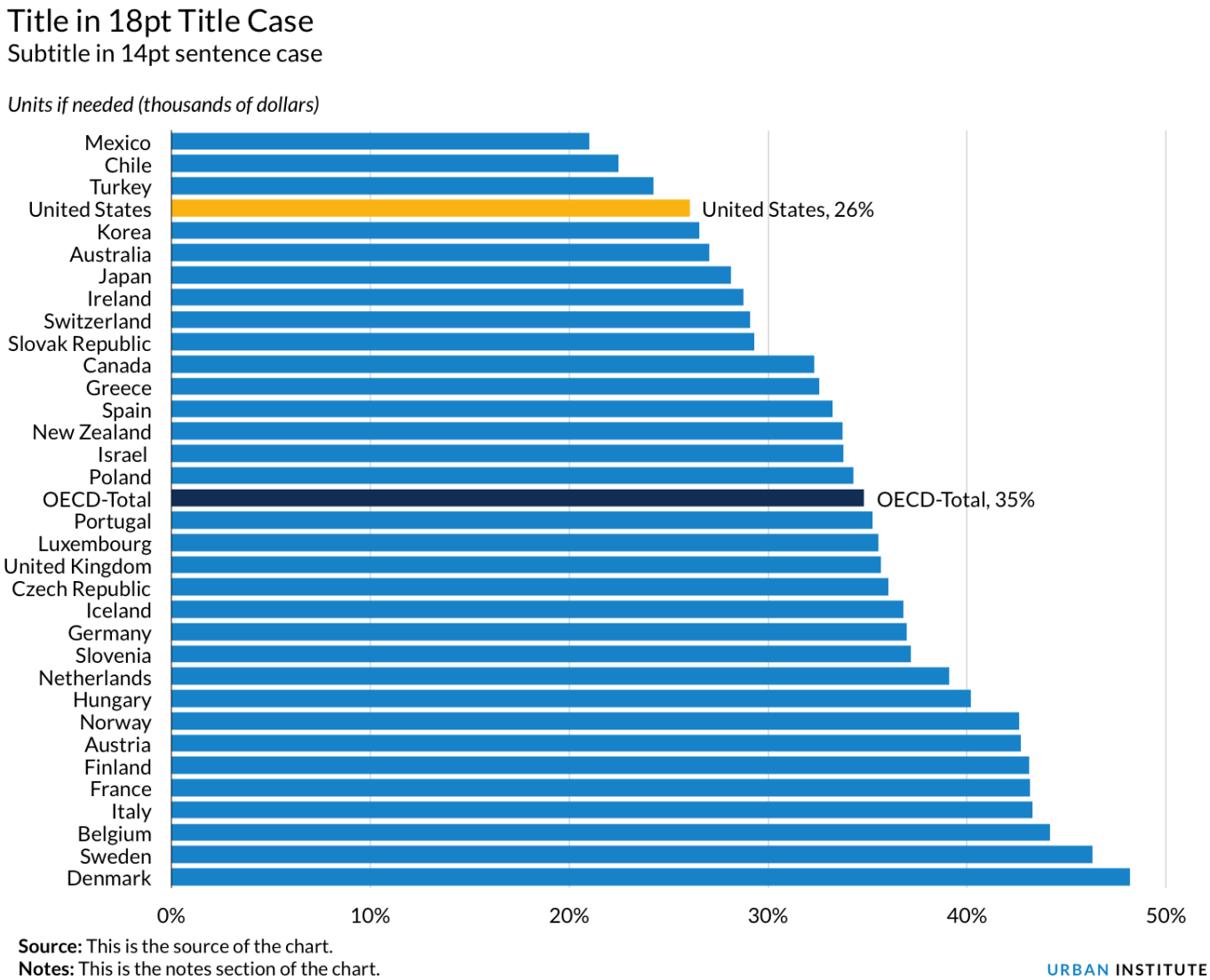
BAR CHARTS

WHEN TO USE A BAR CHART

- To show the trend in one variable, usually across a number of categories.
- To show multiple variables with multiple bars(if they are on the same scale).
- To show the same variable for multiple observations with multiple lines.

STYLE TIPS

- The y-axis should start at zero (there are a few instances when it is okay for the y-axis not to start at zero). (http://qz.com/418083/its-ok-not-to-start-your-y-axis-at-zero/)
- Axis labels should always be horizontal. If you have long labels, consider making a horizontal bar chart instead of a column chart.
- When using a horizontal bar chart, right-align the category labels and center them vertically with the respect to the bar.
- Try to avoid vertical grid lines. Instead, directly label each bar.



NOTES ABOUT THIS CHART

- United States is highlighted using the Poppy color. OECD-Total is differentiated by using a darker shade of the primary bar color

AREA AND STACKED COLUMN CHARTS

WHEN TO USE AN AREA CHART

- To show the trend in composition of a group over time.
- To display the part-to-whole relationship of categories, as well as totals.

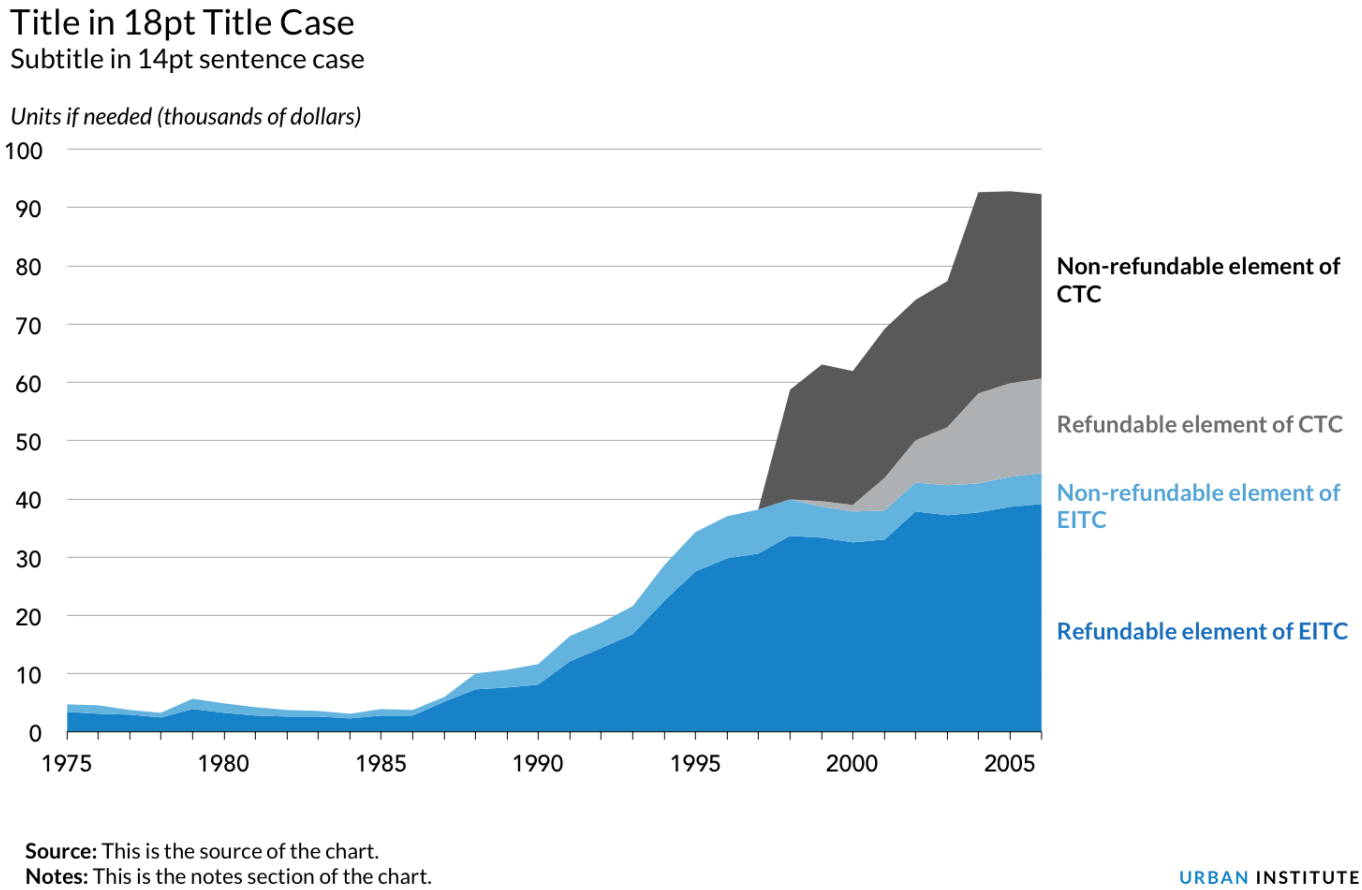
WHEN TO USE A STACKED COLUMN CHART

- When you want to show the composition of groups, but the data is not in a continuous time series.
- A 100 percent stacked area chart can be used when displaying composition over time.

STYLE TIPS

- The y-axis should always* (http://qz.com/418083/its-ok-not-to-start-your-y-axis-at-zero/) start at zero.
- When possible, directly label series. If they are too small, use a legend.
- Avoid individual data labels.

- In a single chart, try to keep the maximum number of categories to three or four. More is not always better. Plotting too many categories on the same chart gives a confusing picture and defeats the purpose.
- Legends should be stretched across the top of the chart, or on the right, and the order should match the order in the chart.
- Sequential series should be shaded from lightest to darkest for easy comparison.

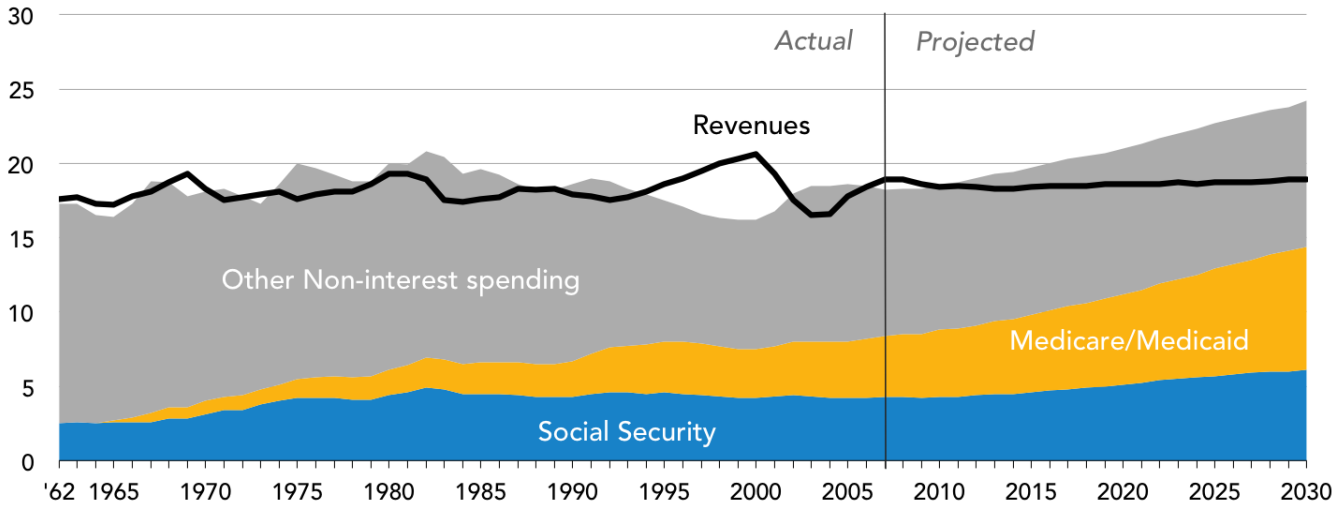


NOTES ABOUT THIS CHART

- This chart is an example of how to use two color families to tell a better story. Here, the CTC series are in shades of gray and the EITC series are in blues. This helps the user visually connect each series.
- Instead of a key, there is enough room to directly label the series using text boxes. Coordinated text color also helps the user make the connection between the label and the series, but beware of colors that are very light or otherwise difficult to read when applied to text.

Title in 18pt Title Case
Subtitle in 14pt sentence case

Units if needed (thousands of dollars)



Source: This is the source of the chart.
Notes: This is the notes section of the chart.

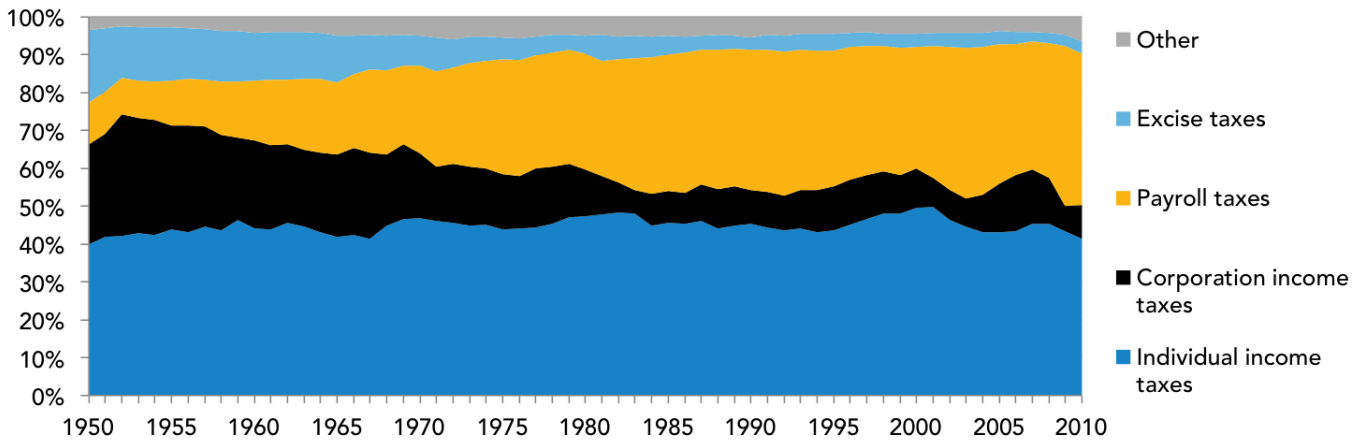
URBAN INSTITUTE

NOTES ABOUT THIS CHART

- X-axis labels should be every 5 or 10 years for time-series data like this, but the chart data starts at 1962. To hack this, I created a new column called "chart-year" and only included the years I wanted to show. Because of spacing issues, I had to abbreviate the first year, 1962, as '62.
- It appears that the growth in Medicare and Medicaid spending is a big story in this chart. For that reason, I chose to use the brightest color in the categorical palette for that series.
- Contrast in typographic elements helps establish hierarchy. The "actual" and "projected" labels are different from the series labels, so I made them gray and italic.

Title in 18pt Title Case
Subtitle in 14pt sentence case

Units if needed (thousands of dollars)



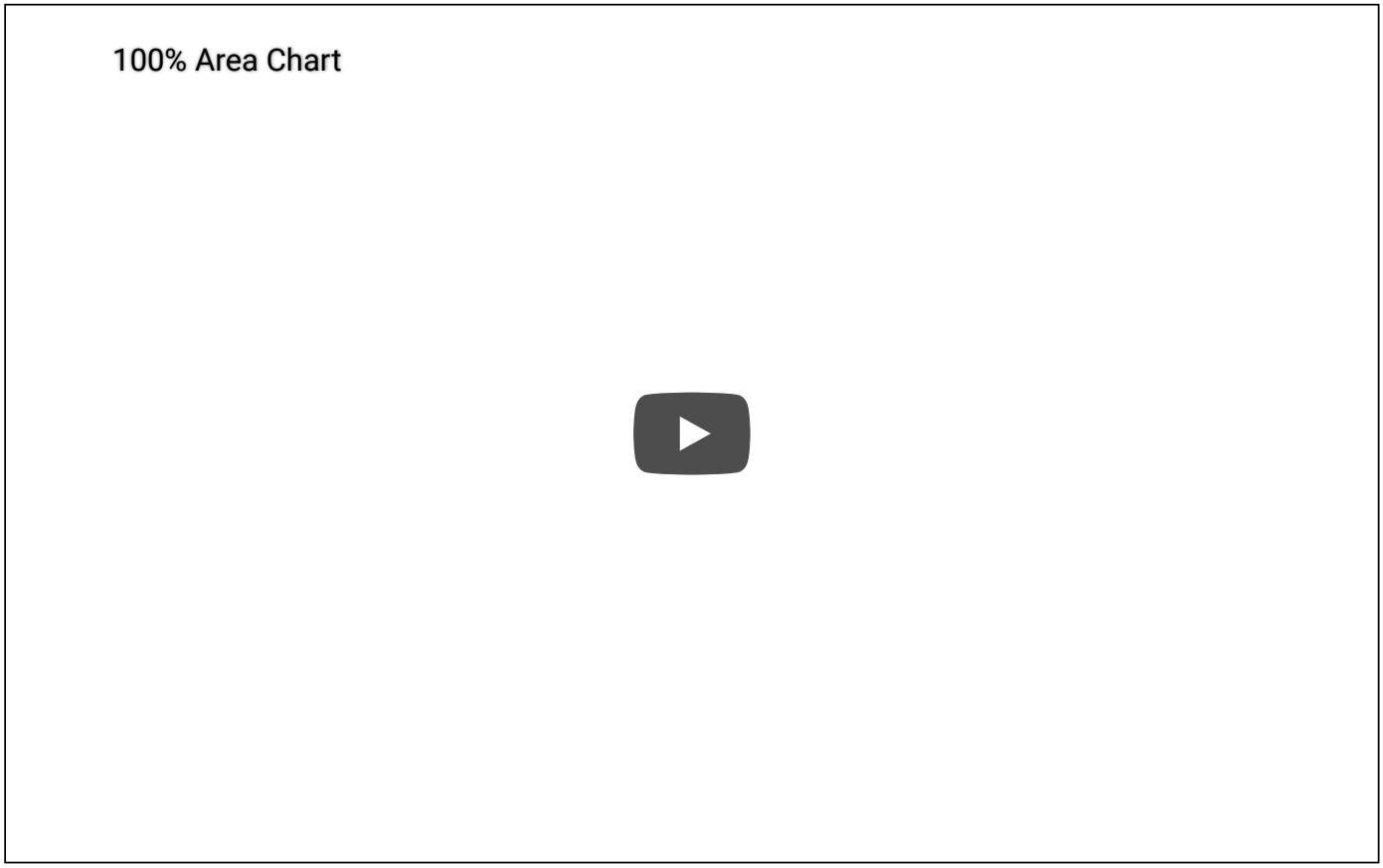
Source: This is the source of the chart.
Notes: This is the notes section of the chart.

URBAN INSTITUTE

NOTES ABOUT THIS CHART

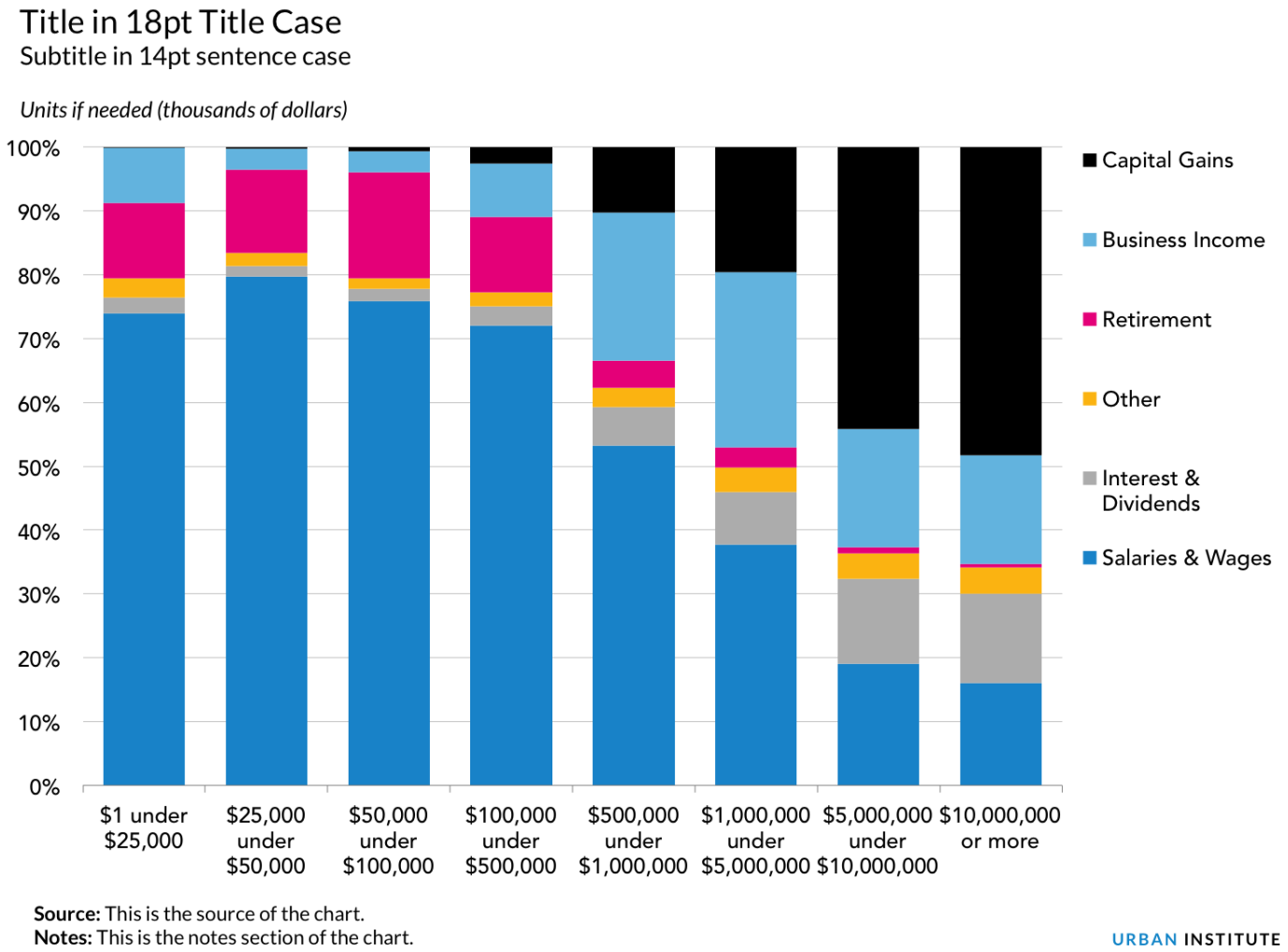
- The x-axis is starting to get a little cramped. To free up some space, we could modify the data so that anything over one million dollars is represented as $1m instead of $1,000,000.
- I put the legend to the right of the chart in a vertical layout. The order matches the order of the data in the columns.

HOW IT'S MADE



CASE STUDY: ALTERNATIVE TO A STACKED COLUMN CHART

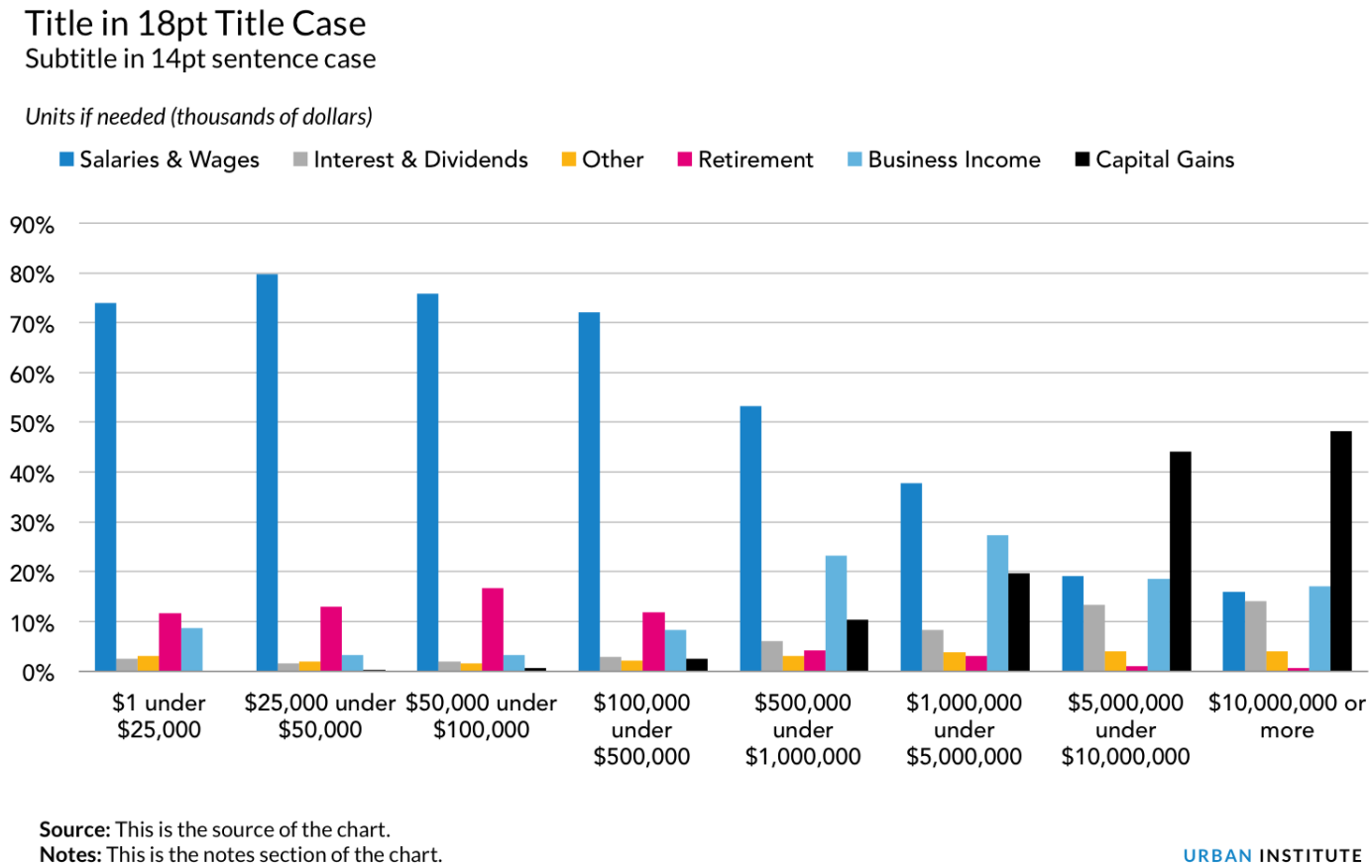
Often, when you want to show composition across several dimensions (usually time), the default choice is a stacked column chart. Though this is not a bad chart type, often a story can be better told by using a different chart.



NOTES ABOUT THIS CHART

- It is hard to see change in the middle series because of the shifting height of the salaries and wages bars.
- The x-axis is starting to get a little cramped. To free up some space, we could modify the data so that anything over one million dollars is represented as $1m instead of $1,000,000.
- Because some series are too small to directly label them, a legend is necessary. The legend is positioned to the right of the chart in a vertical layout. The order matches the order of the data in the columns.

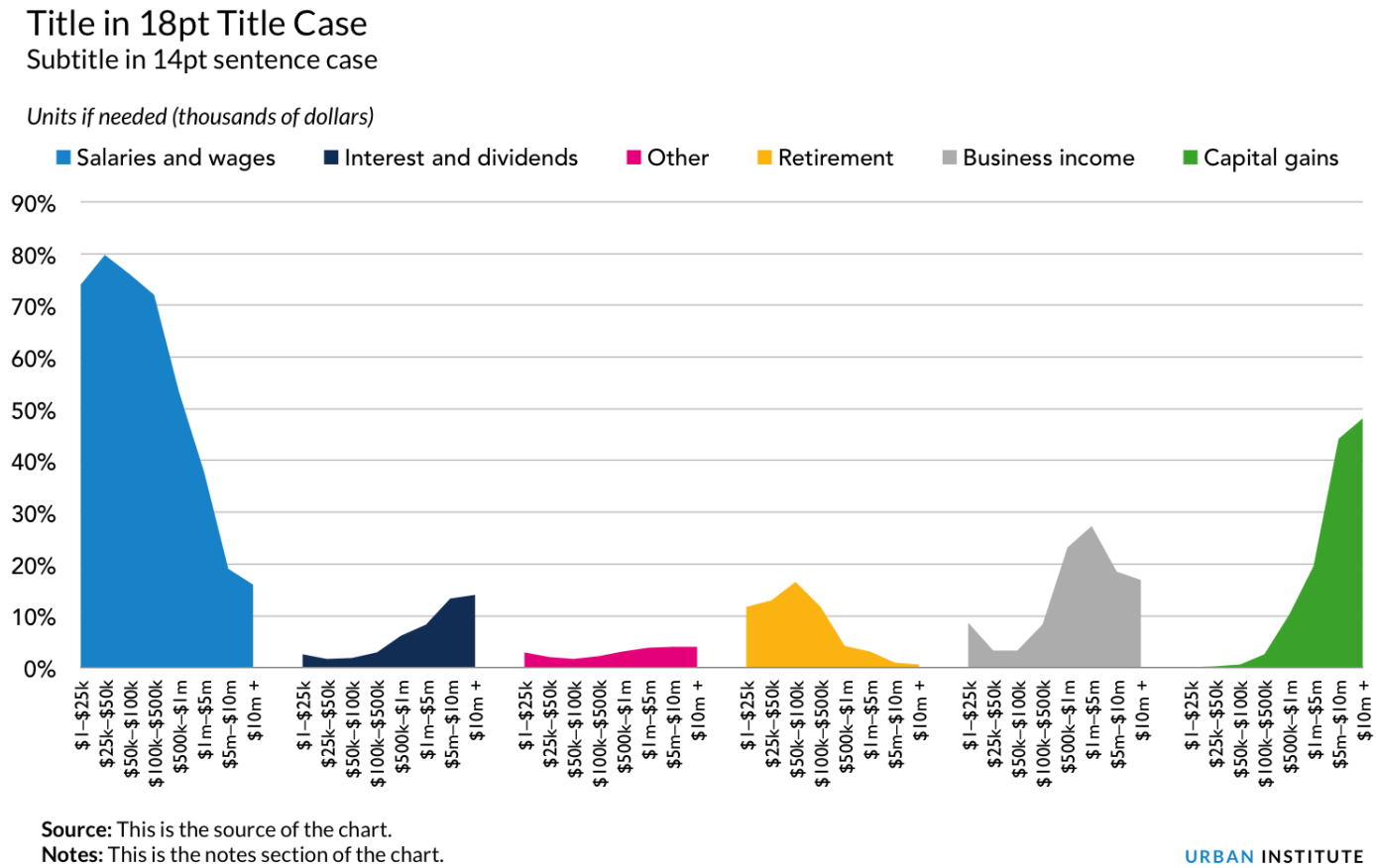
GROUPED COLUMN CHART



NOTES ABOUT THIS CHART

- This version makes it easier to see the change in individual series across categories.
- It is also easier to compare series within the same category, allowing a better visual understanding of the part-to-whole relationship of any one data point.
- The x-axis is starting to get a little cramped. To free up some space, we could modify the data so that anything over one million dollars is represented as $1m instead of $1,000,000.

SMALL MULTIPLES AREA CHARTS



NOTES ABOUT THIS CHART

- If the story for this chart is about the distribution of any given series, small multiple area charts are the way to go.
- For example, it's easy to see that the lower-earning groups get most of their income from salary, whereas the highest-earners get more from capital gains.
- The x-axis labels are a little small.
- The legend is likely unnecessary—the same labels could be accomplished with text boxes.
- The colors here are also unnecessary—There's no need to code the data by color because they're all separate. I do, however, like the contrast the colors give each category.

LINE CHARTS

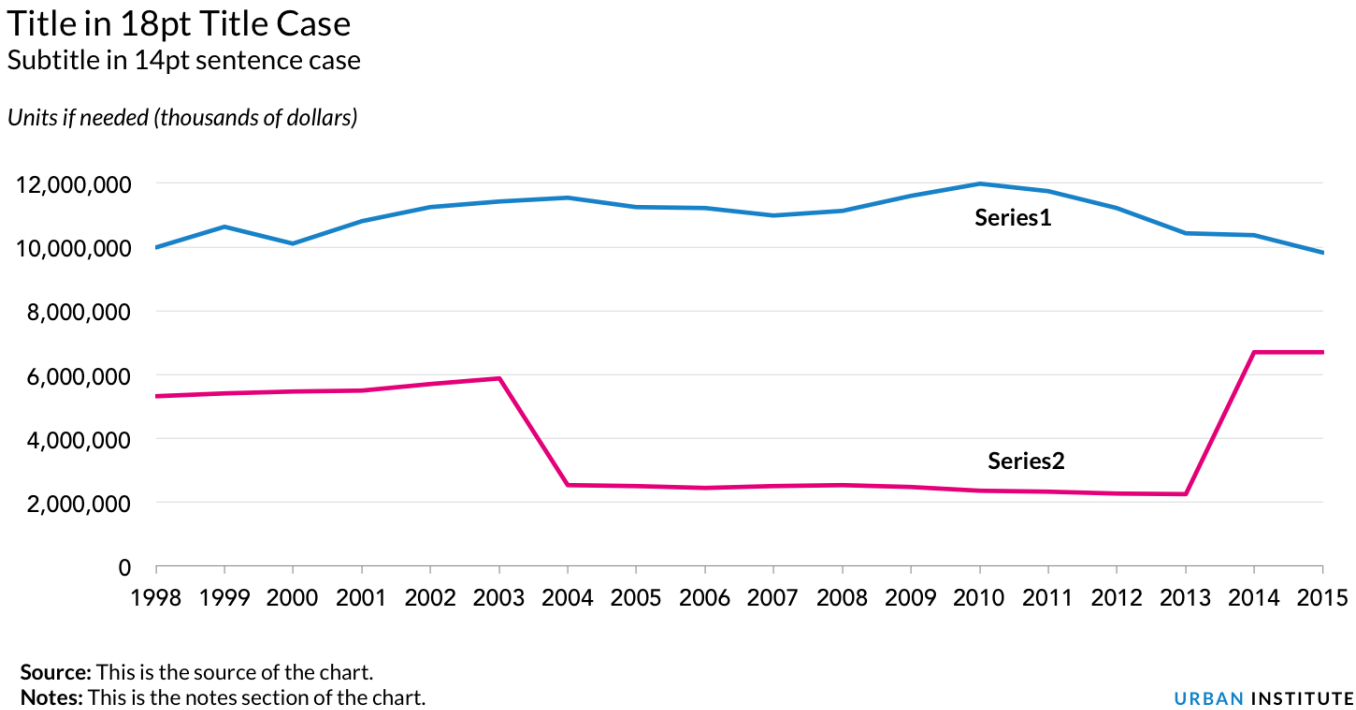
WHEN TO USE A LINE CHART

- To show the trend in one variable, usually over time.
- To show multiple variables with multiple lines (if they are on the same scale).
- To show the same variable for multiple observations with multiple lines.

STYLE TIPS

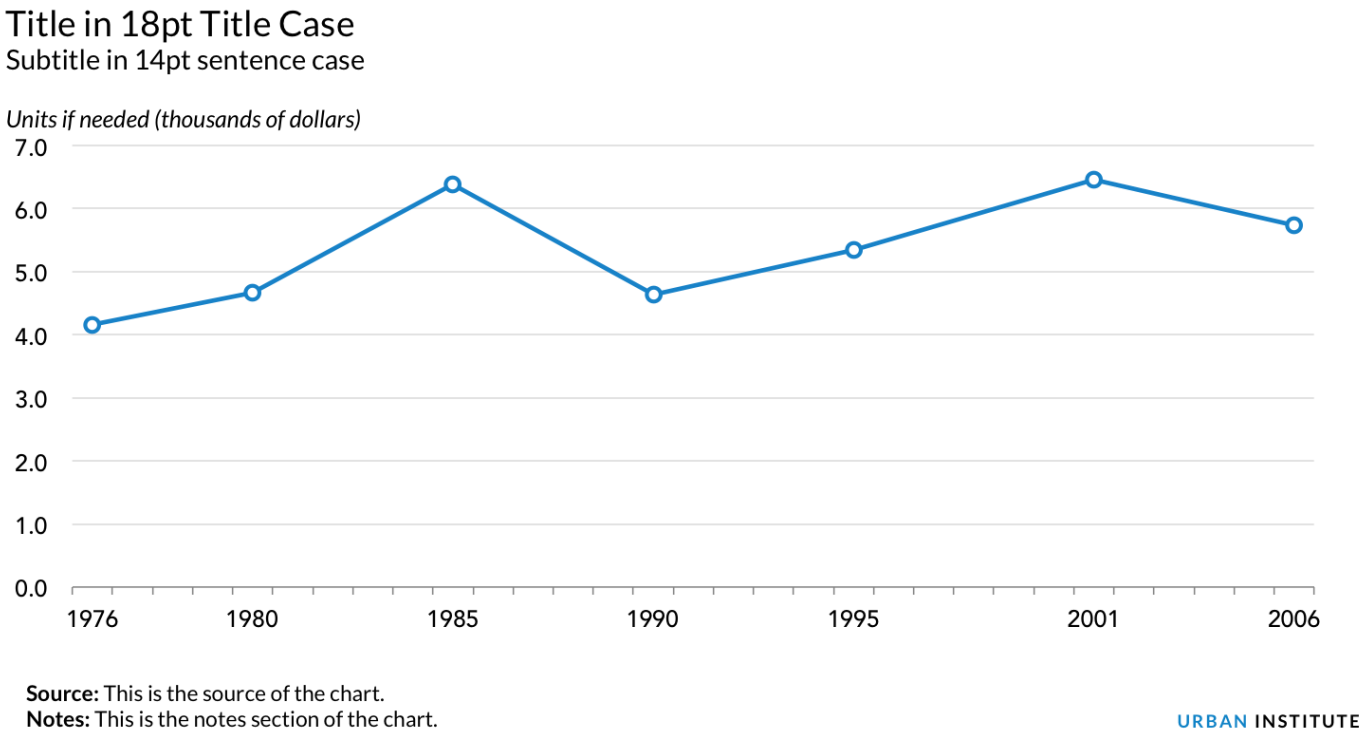
- The y-axis should always* (http://qz.com/418083/its-ok-not-to-start-your-y-axis-at-zero/) start at zero.
- When possible, directly label series. If too close together, use a legend.
- Avoid individual data markers.
- Avoid dashed lines.

- In a single chart, try to keep the maximum number of lines to three or four. More is not always better. Plotting too many lines on the same chart gives a confusing picture and defeats the purpose.
- Legends should be stretched across the top of the chart and the order should match the order in the chart.
- Sequential series should be shaded from lightest to darkest for easy comparison.



NOTES ABOUT THIS CHART

- Supposing the data in this chart is categorical, I used two contrasting colors. If the data were sequential, I would have used two shades of teal.
- The x-axis is starting to get a little cramped. To free up some space, we could reduce the number of labels (labeling every 2, 5, or 10 years) or change the year format to be '98, '99, '00, etc.
- If the labels contain the month and year, the standard format would be "Jan. 2010", for instance. Alternatively, other formats could be "1/2010" or "1/'10".



NOTES ABOUT THIS CHART

- Sometimes, we need to display data that are not on regular intervals. In these cases, it is helpful to mark the data points with individual markers.

PIE AND DISTRIBUTION CHARTS

WHEN TO USE A PIE CHART

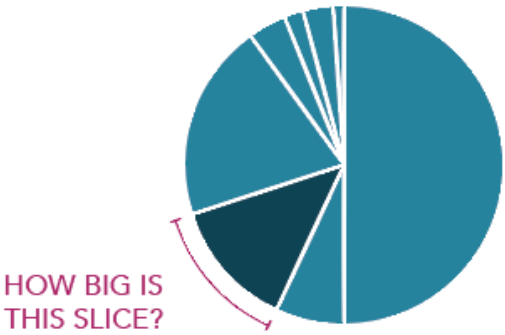
- Use them sparingly. Often a bar or column chart is better. It is difficult to visually judge the size of circles (or circle segments). These segments are easier to discern as rectangles on a scale.
- Use pie charts when you want to show the relative relationship between two or three things
- When they add up to 100 percent (which may necessitate the inclusion of a category such as “none”, or “other”).

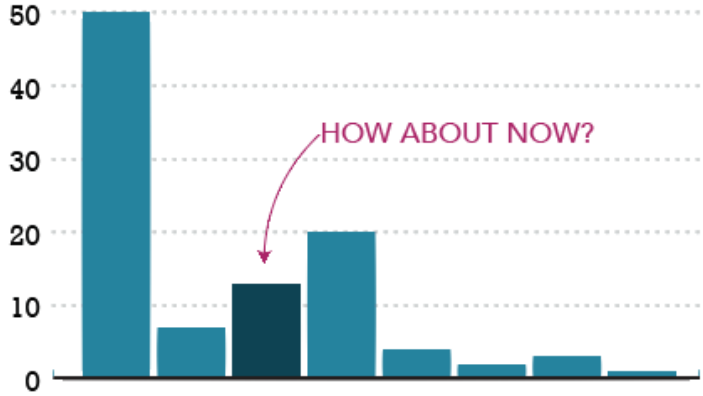
CAUTIONS ABOUT PIE CHARTS

EXCERPTED FROM JON SCHWABISH'S AN ECONOMIST’S GUIDE TO VISUALIZING DATA (HTTP://PUBS.AEAWEB.ORG/DOI/PDFPLUS/10.1257/JEP.28.1.209)

The debate concerning the effectiveness of pie charts is among the most contentious in the field of data visualization. Many people love pie charts—they are familiar, easily understood, and present “part-to-whole” relationships in an obvious way. But others argue that because pie charts force readers to make comparisons using the areas of the slices or the angles formed by the slices—something that our visual perception does not accurately support—they are not an effective way to communicate information.

Pie chart slices that form 90-degree right angles—that is, slices that form one-quarter increments—are the most familiar to our eyes. Other amounts are more difficult to discern. For that reason, a column or bar chart is preferable in most cases.





PIE CHART RESEARCH

- Save the Pies for Desert, (http://www.perceptualedge.com/articles/visual_business_intelligence/save_the_pies_for_dessert.pdf) Stephen Few
- An Economist's Guide to Visualizing Data (http://pubs.aeaweb.org/doi/pdfplus/10.1257/jep.28.1.209), Jon Schwabish
- Pie Charts are the Worst, (http://www.businessinsider.com/pie-charts-are-the-worst-2013-6) Walter Hickey
- Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods, (http://www.jstor.org/stable/2288400?seq=15#page_scan_tab_contents) William S. Cleveland and Robert McGill

The current thinking is pretty much summed up by this tweet from data visualization pioneer Edward Tufte:

> Pie chart users deserve same suspicion+skepticism as those who mix up its/it's, there/their.To compare,use little table, sentence, not pies.
>
> — Edward Tufte (@EdwardTufte) January 10, 2013 (https://twitter.com/EdwardTufte/status/289208399621672960)

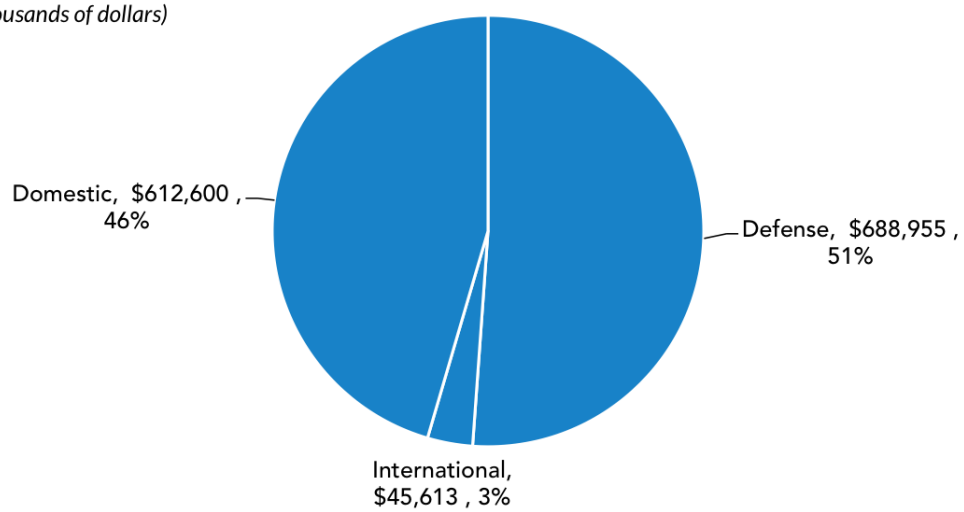
That's not to say that pie charts don't have a place when communicating research. They're OK to use when the number of slices are fewer than four or five— and when you're trying to tell a clear 'part-to-whole' story about a single slice.

PIE CHART EXAMPLES

Title in 18pt Title Case

Subtitle in 14pt sentence case

Units if needed (thousands of dollars)



Source: This is the source of the chart.
Notes: This is the notes section of the chart.

URBAN INSTITUTE

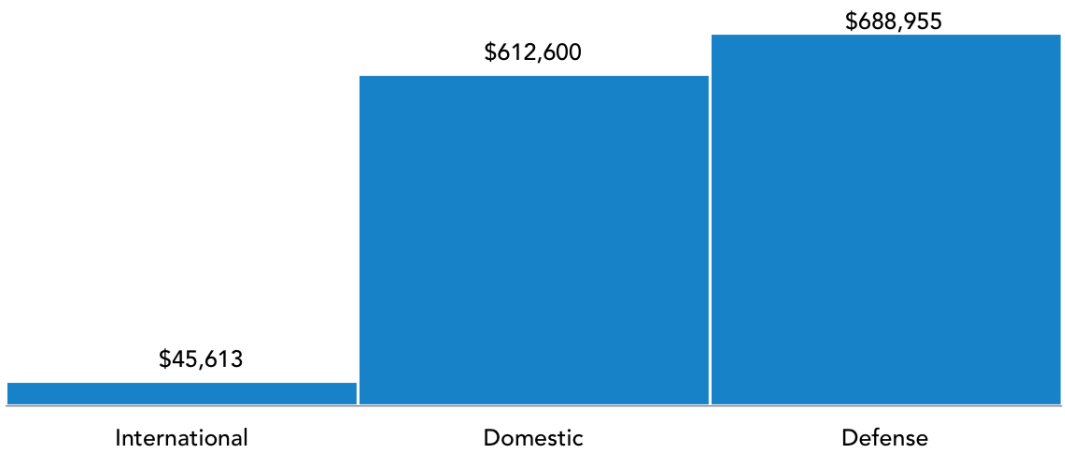
NOTES ABOUT THIS CHART

- Some times, pie charts are useful. In this case, the largest slice is 51%, a value that is very easy to recognize because our eyes can easily find the 50% mark.
- Pie charts work extremely well when they have fewer than four or five slices.
- It's helpful to directly label each slice. Relying on a key requires the reader to do too much work to shift their eyes from one spot to another.
- With only a few colors and direct labels, extra colors become distracting. Stick with only one color and use a highlight if needed.
- An alternative, below, is a histogram. But in this case, the pie chart is nearly as compelling.

Title in 18pt Title Case

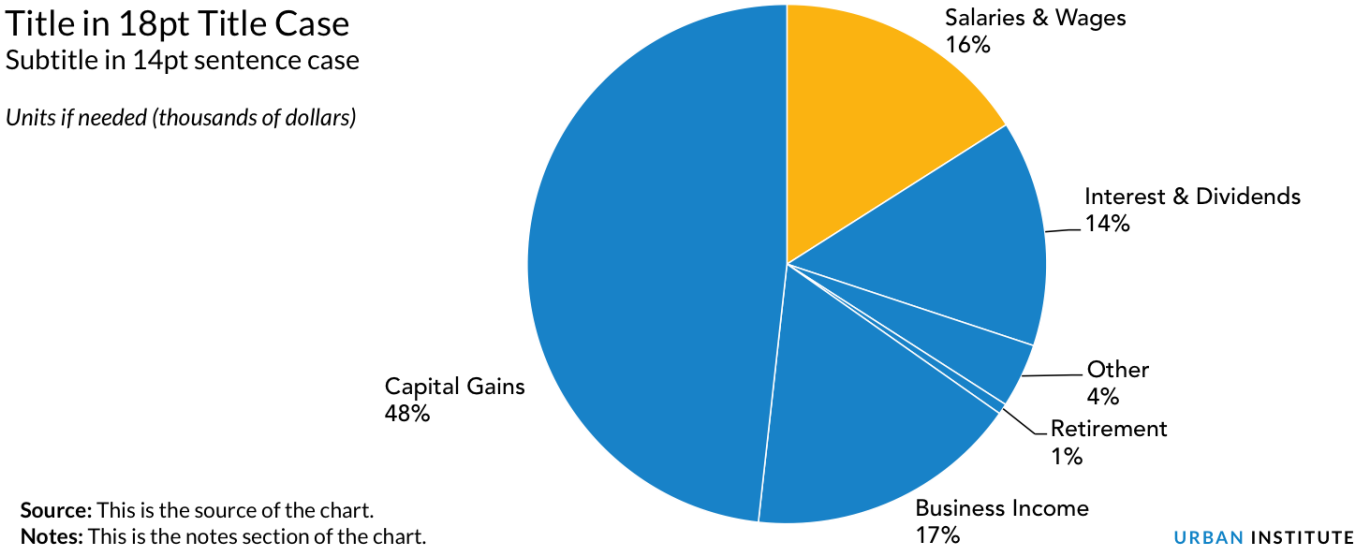
Subtitle in 14pt sentence case

Units if needed (thousands of dollars)



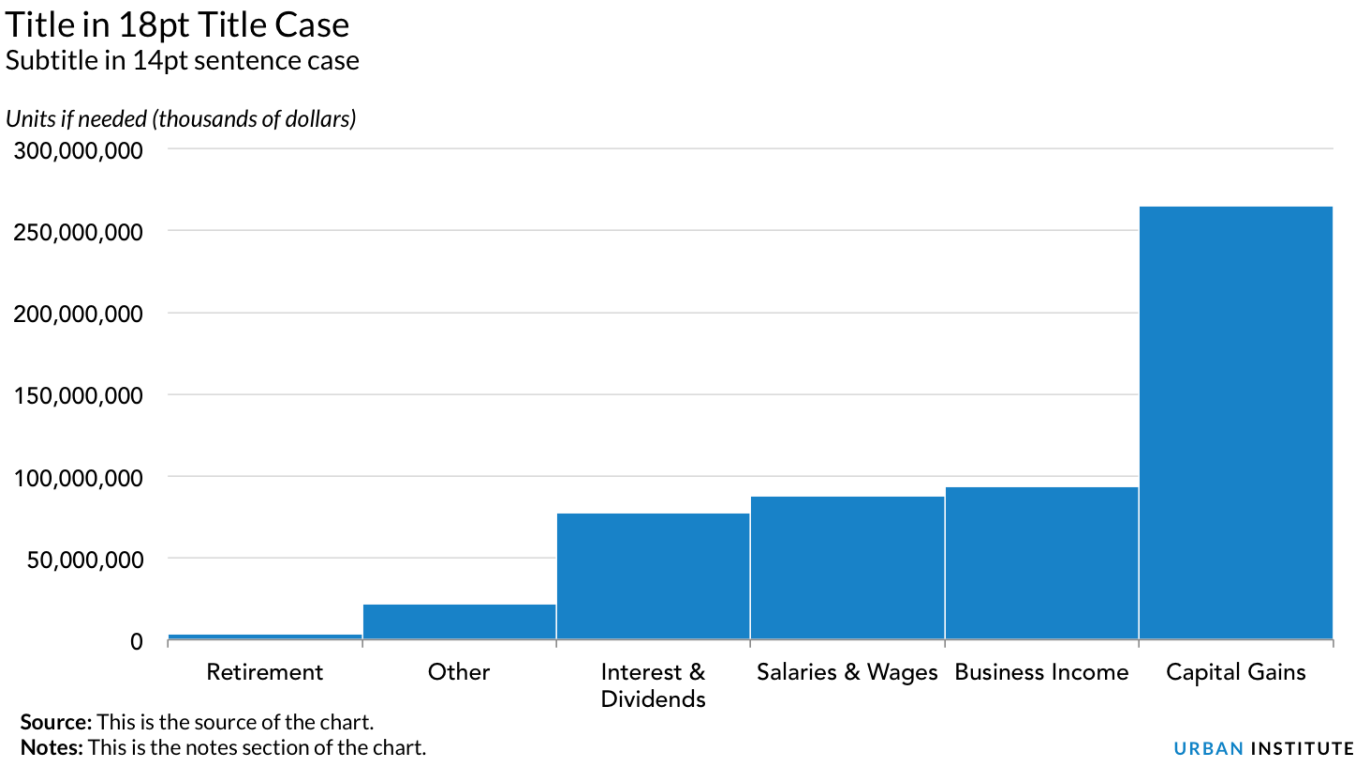
Source: This is the source of the chart.
Notes: This is the notes section of the chart.

URBAN INSTITUTE



NOTES ABOUT THIS CHART

- This pie chart has many more slices than the first example, however It's still fairly easy to make out the individual slices.
- Depending on the purpose of the chart, a histogram may be a better option. If the purpose is to show the size of one slice compared to the rest of the categories, a pie chart makese sense. In that case, it would be good to highlight that individual slice with a different color. If the purpose of the chart is to show how each slice compares to other slices, a histogram would work better (below). Histograms make it much easier to compare values than pie charts.



MAPS

SHOULD IT EVEN BE A MAP?

Just because you've got geographic data, doesn't mean that you have to make a map. (https://drive.google.com/open?id=0B8Ayd1M8yKi4VmE3ejRuTkR3a2c) Many times, there are more efficient storyforms that will get your point across more clearly. If your data shows a very clear geographic trend or if the absolute location of a place or event matters, maps might be the best approach, but sometimes the reflexive impulse to map the data can make you forget that showing the data in another form might answer other—and sometimes more important—questions. (http://www.ericson.net/content/2011/10/when-maps-shouldnt-be-maps/) Consider using other graphic types when the interesting patterns are not geographic patterns, or when the geographic data is more effective for analysis than for presentation. Many times, a simple bar chart, column chart, scatterplot or table are more effective at translating your research to a reader.

In some cases, maps might actually be misleading. State level US maps, for example, give disproportionate weight to states with large land areas, and tend to obscure smaller states in the Northeast. Maps with more granularity can still be misleading. Many times, what appear to be geographic trends are actually directly correlated to population density (wealth, power consumption, etc.), or in some cases directly correlated to some other hidden variable such as racial composition, age distribution, or even weather.

MAP PROJECTIONS

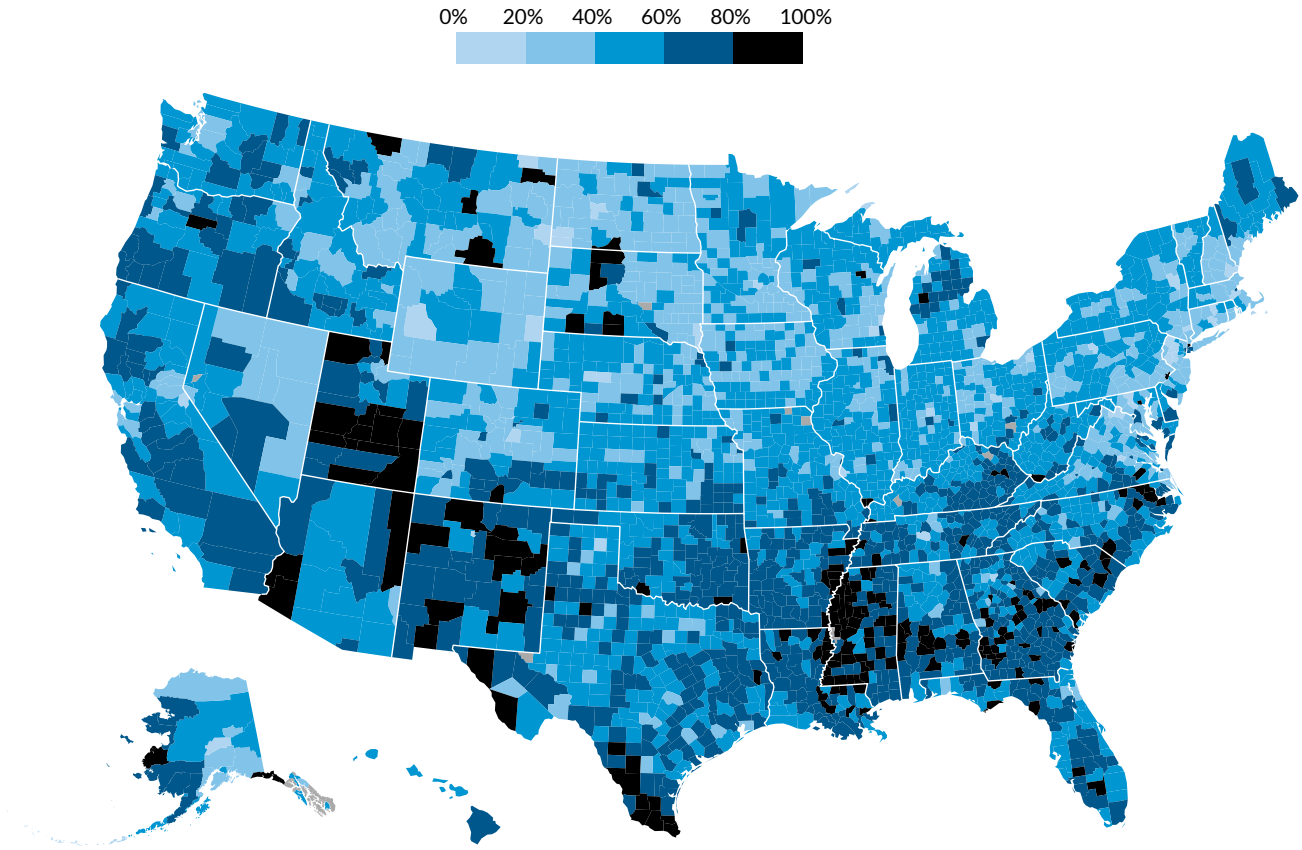
Depending on the purpose and content, maps should use different projections. US maps for print publication, or use in reports should use the Albers Equal Area projection. Simple interactive county- or state-level maps also use the Albers projection. Zoomable tile-based interactive maps (http://apps.urban.org/features/ncdb/immigrants-reshaping-residential-segregation/index.html#) use the Mercator projection.

MAP COLORS

Maps should follow the same basic rules as other graphics when it comes to colors. Choropleth maps (https://en.wikipedia.org/wiki/Choropleth_map), or maps that show the magnitude of a variable, should normally use the blue sequential ramp.

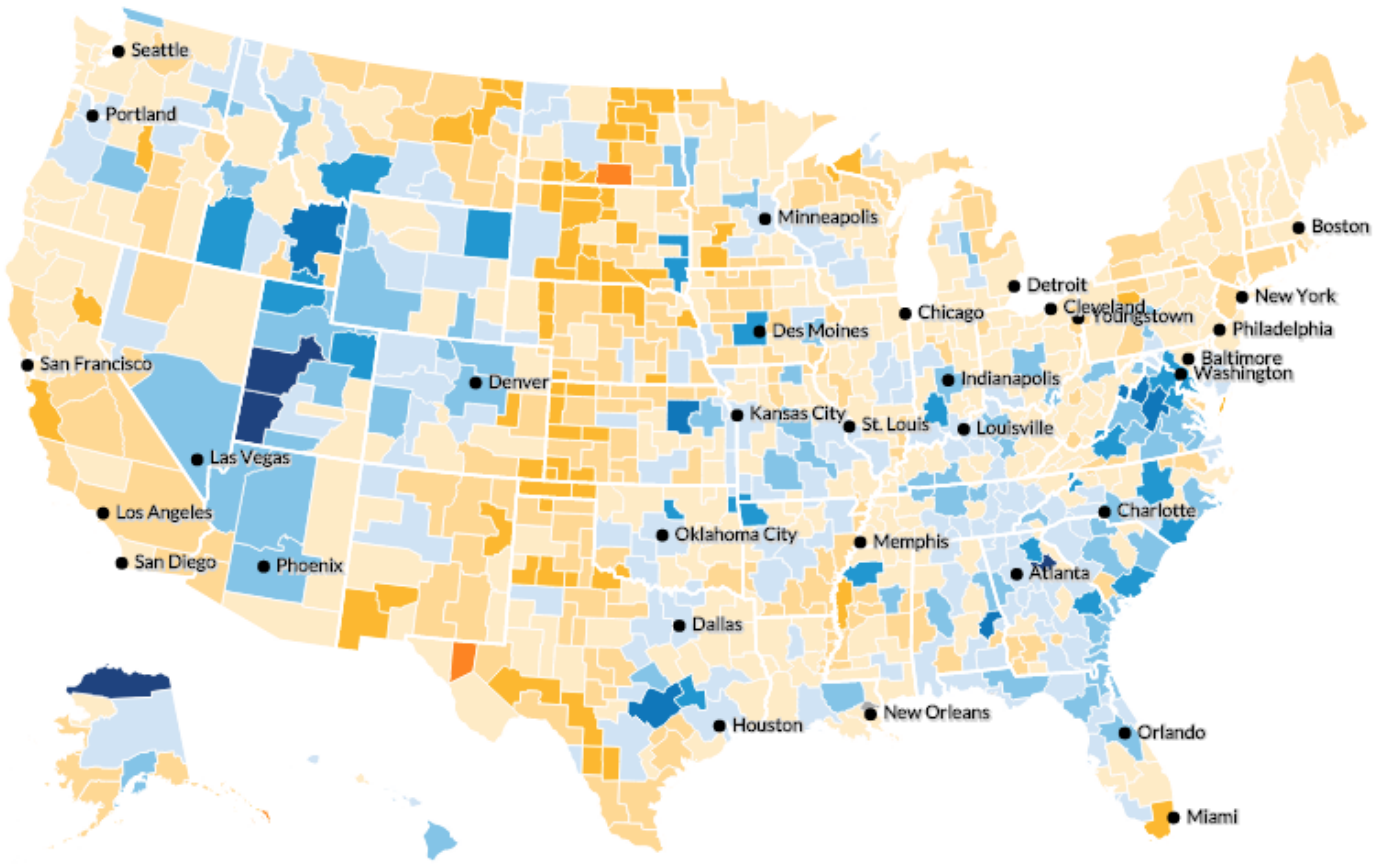
Hex: #cfe8f3	Hex: #73bfe2	Hex: #1696d2	Hex: #0a4c6a	Hex: #000000
rgb(207,232,243)	rgb(115,191,226)	rgb(22,150,210)	rgb(10,76,106)	rgb(0,0,0)
CMYK: 15,5,0,5	CMYK: 49,15,0,11	CMYK: 90,29,0,18	CMYK: 91,28,0,58	CMYK: 0,0,0,100

Share of children from low-income families in public schools



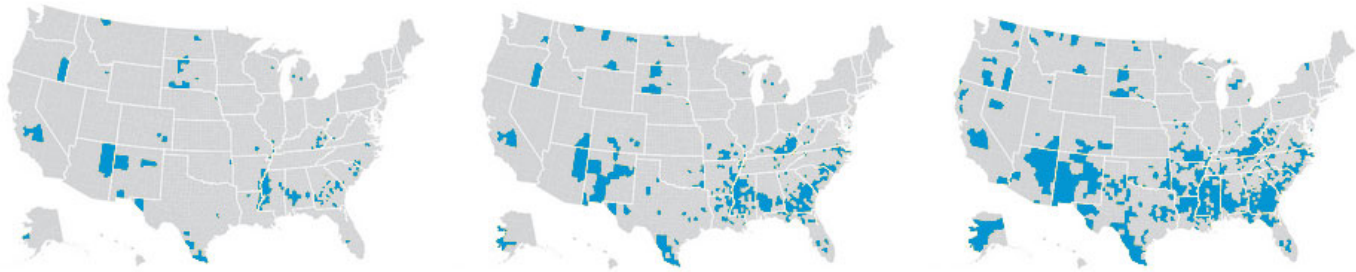
Sometimes, a map needs to display change in a positive and negative way. (http://apps.urban.org/features/mapping-americas-futures/) In those cases, a diverging scheme works best.

Hex: #ca5800 rgb(202,88,0) CMYK: 0,56,100,21	Hex: #fdbf11 rgb(253,191,17) CMYK: 0,25,93,1	Hex: #fdd870 rgb(253,216,112) CMYK: 0,15,56,1	Hex: #fff2cf rgb(255,242,207) CMYK: 0,5,19,0	Hex: #cfe8f3 rgb(207,232,243) CMYK: 15,5,0,5	Hex: #73bfe2 rgb(115,191,226) CMYK: 49,15,0,11	Hex: #1696d2 rgb(22,150,210) CMYK: 90,29,0,18	Hex: #0a4c6a rgb(10,76,106) CMYK: 91,28,0,58



OTHER WAYS TO MAP

Need to show data across multiple points in time? Consider using small map multiples to show the progression or spread.



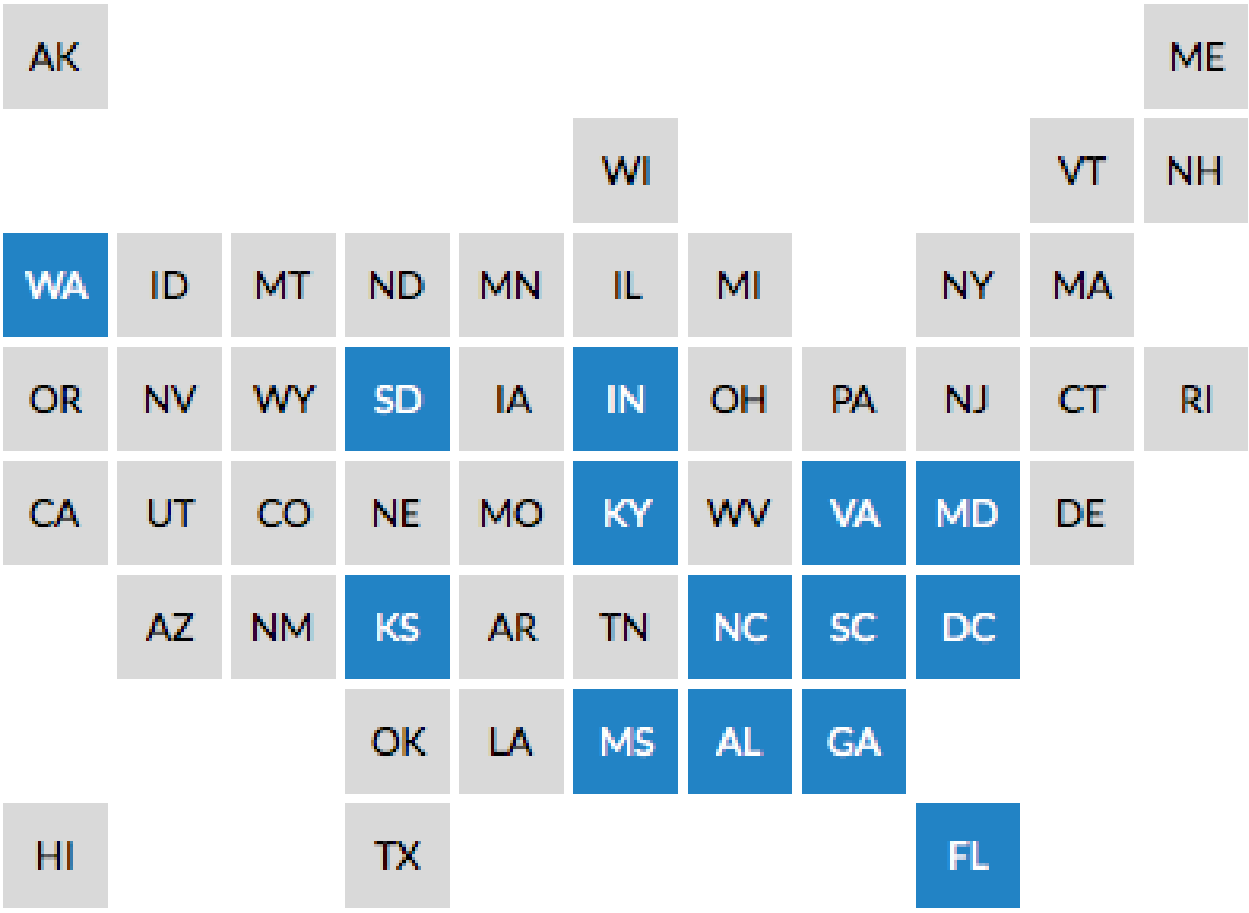
TILE MAPS

The problem with normal state-by-state maps is that the western US has very large states and the eastern US has small states. This leads to the western states being over represented with more visual weight than the states in the northeast.

Recently, many organizations (http://blog.apps.npr.org/2015/05/11/hex-tile-maps.html) have adopted use of grid or tile maps. The advantage of these maps is that each state has equal visual weight.

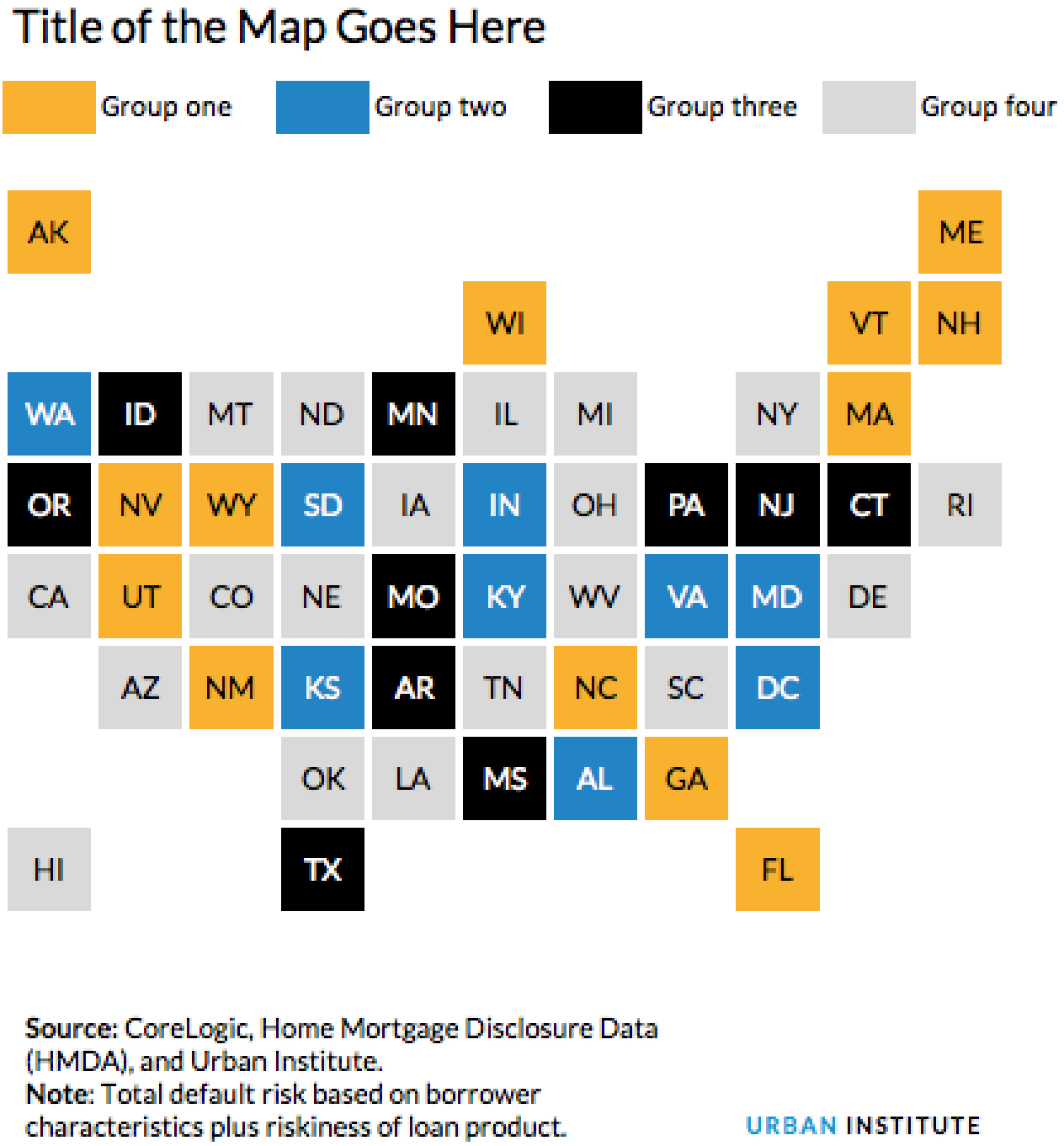
(tile-maps_template.xlsx)Even better, these maps can be created with a simple excel file. (tile-maps_template.xlsx)

Title of the Map Goes Here



Source: CoreLogic, Home Mortgage Disclosure Data (HMDA), and Urban Institute.
Note: Total default risk based on borrower characteristics plus riskiness of loan product.

URBAN INSTITUTE



ADDITIONAL RESOURCES

Have a question or looking for more help? Contact Ben Chartoff (mailto:bchartoff@urban.org) or Jon Schwabish (mailto:jschwabish@urban.org).

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